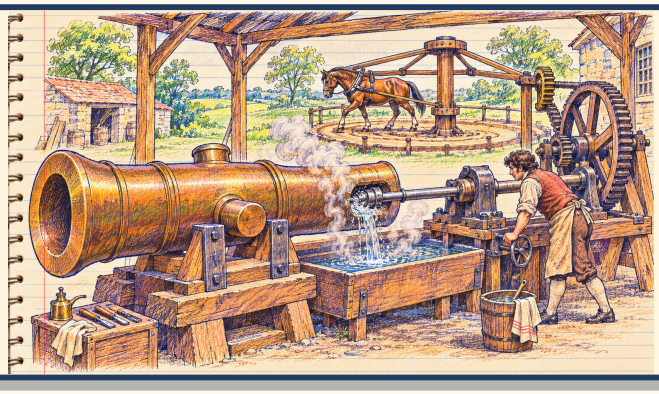
THERMODYNAMICS.

Heat. Work. Change.

The complete teaching script, brought to life with illustrated apparatus, colourful physics notes, and properly typeset mathematics.

Demo Script — Draft for approval
Dr. Soumya Ranjan Senapati



38	13	31	05
source pages	worked problems	source figures redrawn	interactive artifacts

▼ **About this illustrated edition**

All narration, question data, worked solutions, sequence, trivia, and source notes are retained. The illustrations follow a hand-drawn science-notebook style, with recognisable objects and coloured-pencil shading. Equations use proper mathematical typesetting. Historical scenes are illustrative reconstructions.

Flow kept from the source notes: Thermodynamic Systems → First Law → Entropy → Calculating Entropy → Second Law → Thermodynamic Processes → Heat Engines → Carnot Cycle. All original questions replaced with new, harder HL-style problems (with full solutions).

The HTML has print styling and expands worked solutions when printed. The PDF companion contains expanded solutions and static illustrations.

00 INTRODUCTION — WHAT IS HEAT?

FIGURE 00

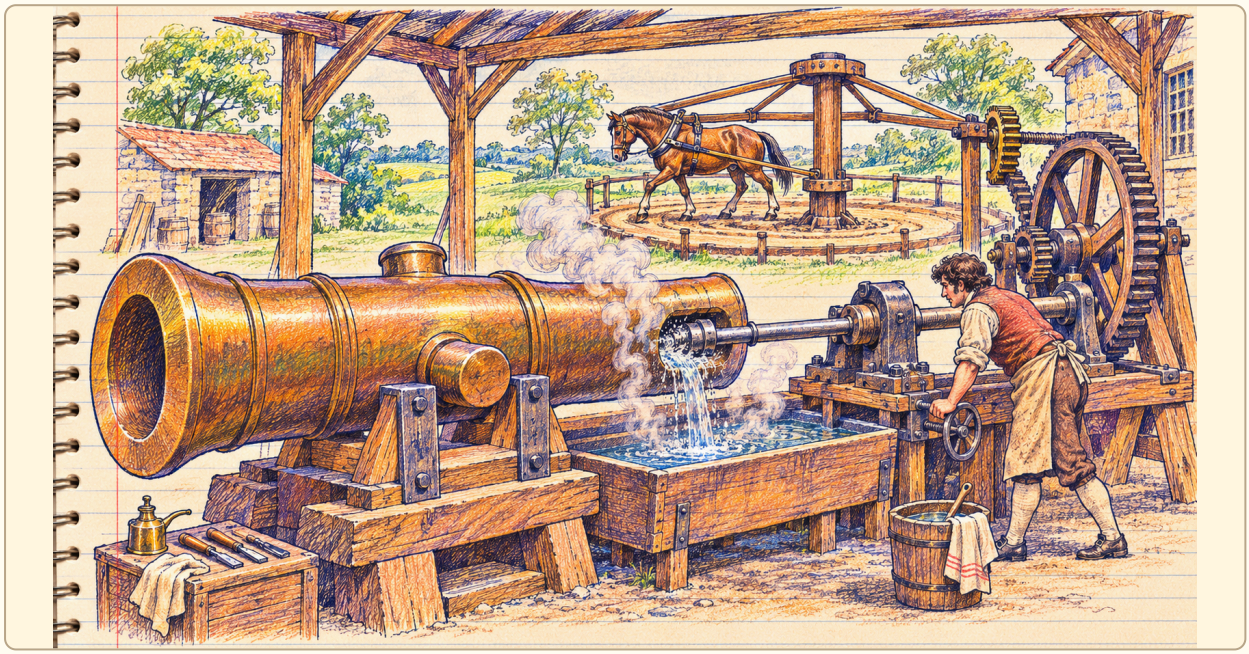


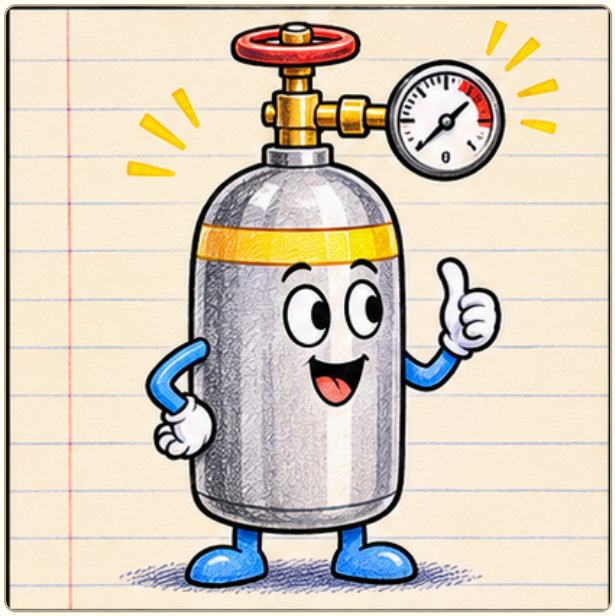
Figure 0. A brass cannon barrel clamped in a Munich workshop, 1798; a blunt drill grinding into it, horses turning the drill, the barrel sitting in a tub of water that is starting to boil.

AI-generated notebook illustration · diagram geometry and question data define the physics.

In 1798, Count Rumford (Benjamin Thompson) saw that boring a cannon barrel kept the surrounding water boiling for as long as the drill turned. Heat was then believed to be a fluid called caloric stored inside objects, so it should have run out. It never did. Rumford concluded that heat is not a substance but the motion of particles. That is exactly what internal energy is: the energy of moving particles. So we start with the particles inside a gas.

01 1. INTERNAL ENERGY

FIGURE 01



PHYSICS MODEL

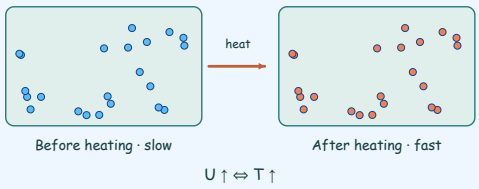
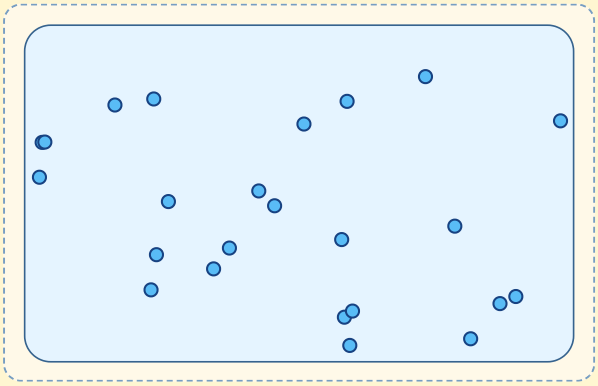


Figure 1. A sealed container of gas on a flame. Left: slow molecules with short arrows. Right (after heating): the same molecules with long arrows.

AI-generated notebook illustration · diagram geometry and question data define the physics.

INTERACTIVE ARTIFACT 01 · PARTICLE MODEL

Turn up the temperature.



Qualitative motion · fixed volume and particle number.

Temperature / K

300 K

$U / U_{300} = 1.00$

Typical speed / $speed_{300} = 1.00$

For the same ideal monatomic gas, internal energy scales with absolute temperature.

Heat a container of gas, and the molecules start moving faster. Faster molecules mean more kinetic energy. In a solid, the molecules can't run around, so they vibrate harder instead. In liquids and solids the molecules are close together and held by intermolecular forces, so they carry both kinetic and potential energy. But an ideal gas is special. We assume its molecules feel no intermolecular forces at all. No forces means no potential energy. So the internal energy of an ideal gas is purely kinetic. That gives us a beautifully simple rule: the change in internal energy depends only on the change in temperature.

$\Delta U \propto \Delta T$ For an ideal monatomic gas:

$$\Delta U = \frac{3}{2} N k_B \Delta T$$

ΔU = change in internal energy (J)

k_B = Boltzmann constant ($1.38 \times 10^{-23}\,JK^{-1}$)

ΔT = change in temperature (K)

N = number of particles Or, in terms of moles:

$$\Delta U = \frac{3}{2} nR\Delta T$$

n = number of moles (mol)

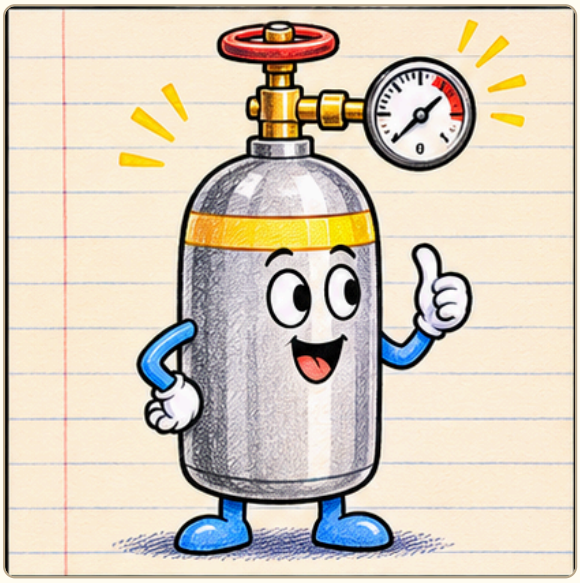
R = molar gas constant ($8.31 Jmol^{-1}K^{-1}$) And since $pV = nRT$, there is a lovely shortcut: the total internal energy of an ideal monatomic gas is simply $U = (\frac{3}{2})pV$. Keep that in your pocket — we'll use it again and again.

Examiner tip: If a question asks for the total internal energy of an ideal gas, that is the same as its total kinetic energy, because the potential energy is zero.

WORKED PROBLEM 01 / 13

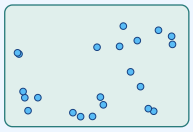
Problem 1 — The Helium Tank

Tsokos (CUP, 3rd ed.) B.4 · Internal energy of an ideal gas · replaces SME: 50 → 150 °C claim & 126 mol expansion



QUESTION DATA & MODEL

RIGID HELIUM TANK



$V = 0.050\ m^3$

$p = 2.0 \times 10^5\ Pa$

$T = 27\ °C$

Double U: what temperature?

Visual setup · The Helium Tank

AI-generated notebook illustration · diagram geometry and question data define the physics.

A rigid steel tank of volume $0.050 m^3$ holds helium (an ideal monatomic gas) at a pressure of 2.0×10^5 Pa and a temperature of 27 °C.

(a) Calculate the total internal energy of the helium.

(b) Calculate the number of helium atoms in the tank.

(c) A student says: "To double the internal energy, heat it to 54 °C." Find the correct temperature in °C.

(d) Determine the energy that must be supplied by heating to do this.

▼ Worked solution

(a) $U = (\frac{3}{2})pV = 1.5 \times (2.0 \times 10^5) \times 0.050 = 1.5 \times 10^4\ J$

$$N = \frac{pV}{(k_B T)}$$

(b) $$= \frac{(2.0\times10^{5}\times0.050)}{(1.38\times10^{-23}\times300)}$$

$$= 2.4\times10^{24}\ \text{atoms}$$

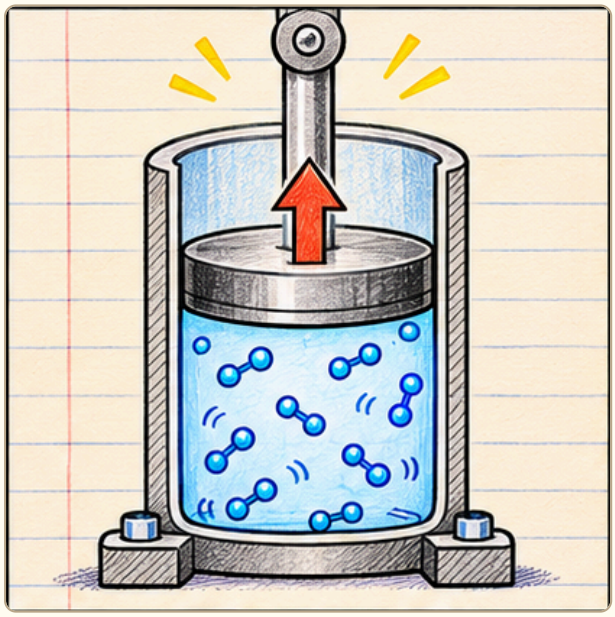
(c) $U \propto T$ in kelvin, not Celsius. 27 °C = 300K. Doubling U means doubling T to 600K = 327 °C. The student doubled the Celsius number, which means nothing physically: 54 °C is only 327 K, a rise of just 9%.

(d) The tank is rigid, so no work is done ($W = 0$). All the heat goes into internal energy:

$$Q = \Delta U = 3.0\times10^{4} - 1.5\times10^{4} = 1.5\times10^{4}\ \text{J}$$

02 2. WORK DONE BY A GAS

FIGURE 02



PHYSICS MODEL

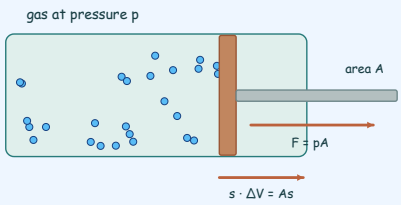


Figure 2. A gas in a cylinder with a piston of area A. The gas pushes the piston out a distance s. Force $F = pA$ shown on the piston.

AI-generated notebook illustration · diagram geometry and question data define the physics.

A gas that expands doesn't just sit there. It pushes. It does work on its surroundings. Push the piston in from outside, and the reverse happens: work is done on the gas. At constant pressure, the force on the piston is

$p = \frac{F}{A} \Rightarrow F = pA$ Work is force times displacement, $W = Fs$, so $W = pAs$. And $A \times s$ is just the change in volume, ΔV:

$$W = p\Delta V$$

W = work done (J)

p = pressure (Pa)

ΔV = change in volume (m^3) This assumes the surrounding pressure stays constant while the gas expands — true, for example, when a gas expands against the atmosphere, whose pressure changes very slowly.

Examiner tip: The p in $W = p\Delta V$ is really the pressure of the surroundings the gas pushes against, because the work is done on the surroundings.

p–V diagrams

FIGURE 03

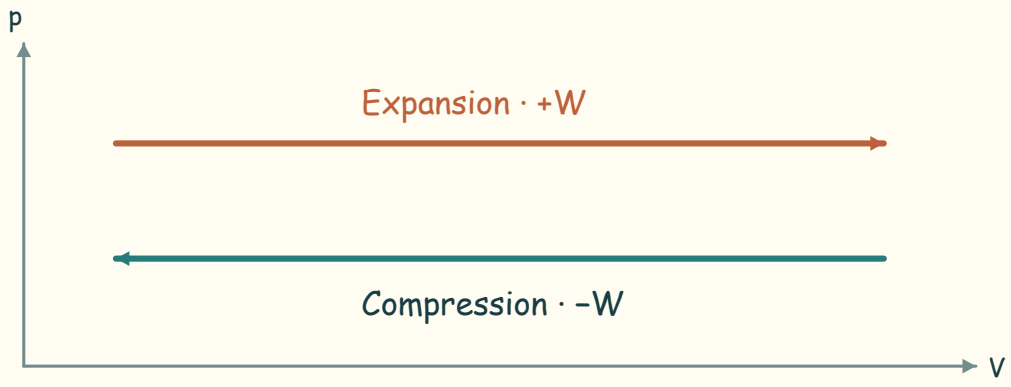


Figure 3. Two p-V graphs. Left: horizontal arrow to the right (expansion), shaded rectangle, "+W, work done BY gas". Right: horizontal arrow to the left (compression), shaded rectangle, "−W, work done ON gas".

Expansion: volume increases ($+\Delta V$), work done by the gas, +W. Compression: volume decreases ($-\Delta V$), work done on the gas, −W.

FIGURE 04

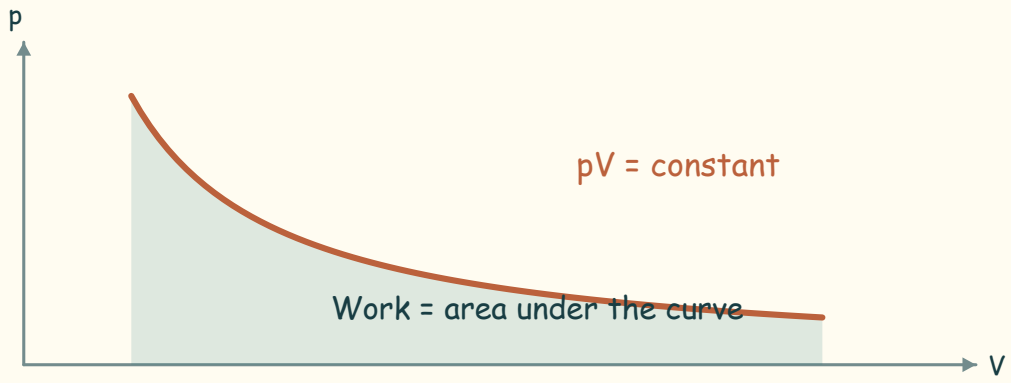


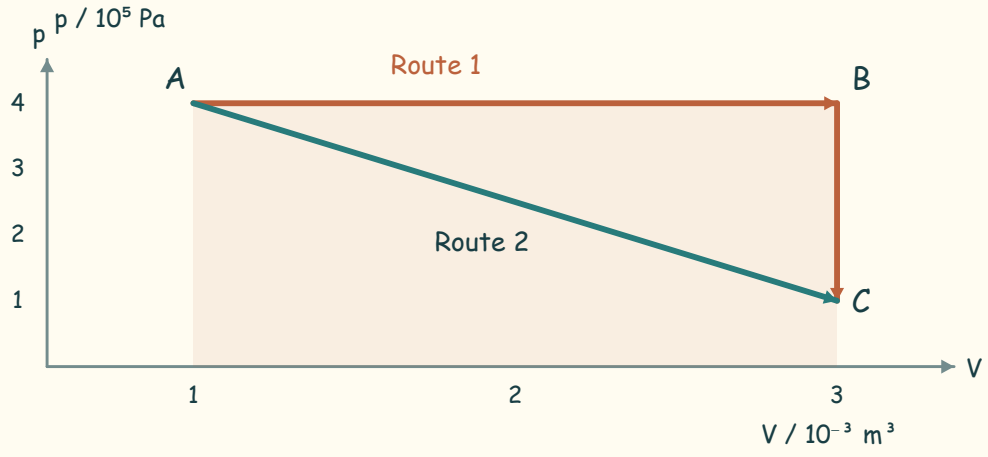
Figure 4. A curved path on a p-V diagram with the area under it shaded.

And when both pressure and volume change at the same time? The rule is universal: work done = area under the p-V graph. Because work is an area, two different paths between the same two states can give different work. The next problem checks this.

WORKED PROBLEM 02 / 13

Problem 2 — Two Paths, Same End States

Tsokos (CUP) B.4 · Work as area under p-V graph · replaces SME: balloon & triangular p-V area



Visual setup · Two Paths, Same End States

An ideal monatomic gas is taken from state A (1.0×10^{-3} m³, 4.0×10^{5} Pa) to state C (3.0×10^{-3} m³, 1.0×10^{5} Pa) by two different routes: Route 1: A → B at constant pressure to $3.0 \times 10^{-3}\ m^3$, then B → C at constant volume. Route 2: A straight line on the p-V diagram directly from A to C.

FIGURE 05

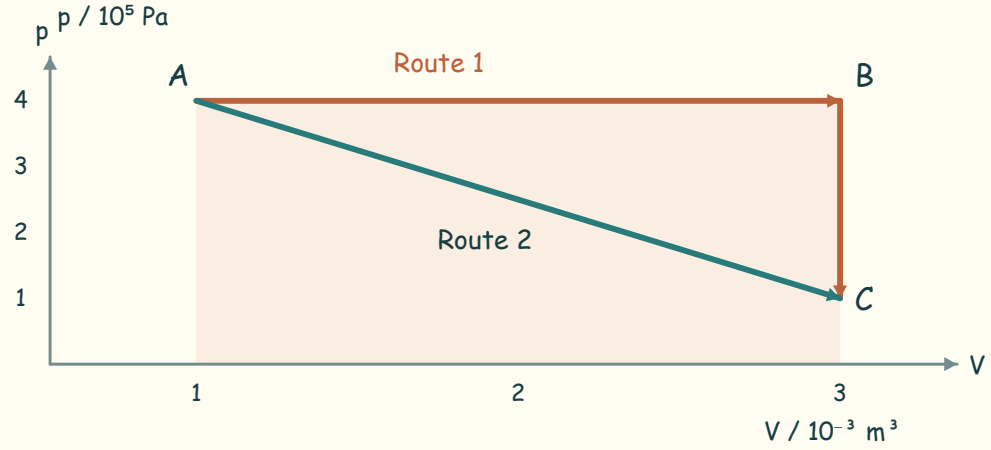


Figure 5. p–V diagram with A (top left), B (top right), C (bottom right). Route 1 = along the top, then down. Route 2 = diagonal from A to C.

(a) Calculate the work done by the gas along Route 1.

(b) Calculate the work done by the gas along Route 2.

(c) The gas is then returned from C to A along the straight line. State the work done and its sign.

▼ Worked solution

(a) $A \rightarrow B$: $W = p\Delta V = (4.0 \times 10^5)(2.0 \times 10^{-3}) = 800\,\text{J}$ $B \rightarrow C$: constant volume, $W = 0$

$W_1 = +800\,\text{J}$

(b) Area of a trapezium = average pressure $\times \Delta V = \frac{1}{2}(4.0 \times 10^5 + 1.0 \times 10^5)(2.0 \times 10^{-3}) = +500\,\text{J}$

(c) Same area, but volume decreases, so work is done on the gas: $W = -500\,\text{J}$ Same start and end states, different work: work depends on the path.

03 3. THE FIRST LAW OF THERMODYNAMICS

HEAT WORK INTERNAL ENERGY

FIGURE 06

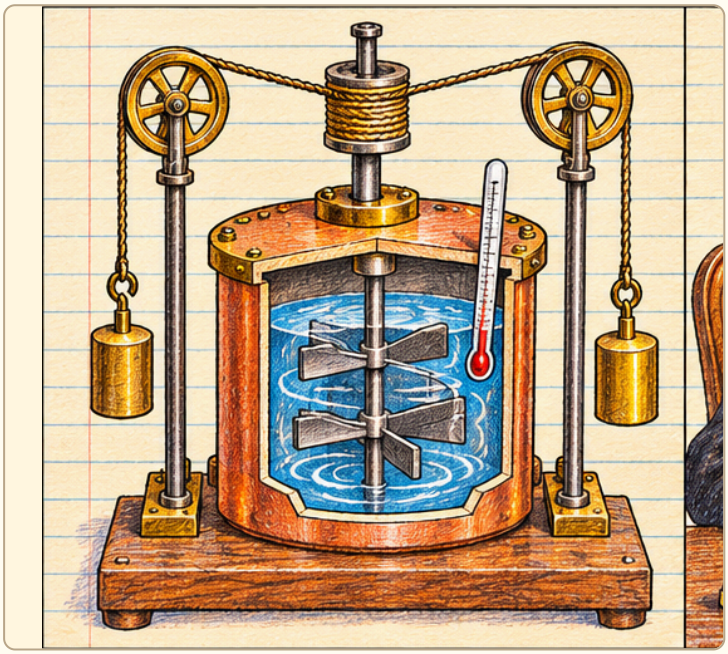


Figure 6. James Joule's paddle-wheel apparatus: falling weights turn paddles in an insulated water tank, a thermometer reads the temperature rise.

AI-generated notebook illustration · diagram geometry and question data define the physics.

In the 1840s, James Joule let falling weights spin paddles in insulated water and measured the temperature rise. A fixed amount of mechanical work always produced the same heating effect. So heat and work are both transfers of energy, and energy is conserved. The unit of energy, the joule, is named after him. Written for a thermal system, this is the First Law of Thermodynamics: Energy supplied by heating = change in internal energy + work done by the system

$$Q = \Delta U + W$$

Q = energy supplied to the system by heating (J)

ΔU = change in internal energy (J)

W = work done by the system (J) It applies to every situation, not just gases. And it comes with a sign convention you must own:

Quantity	Positive (+)	Negative (−)
Q	heat added to the system	heat removed from the system
ΔU	internal energy increases	internal energy decreases
W	work done by the system	work done on the system

Reading p–V graphs with the First Law

FIGURE 07

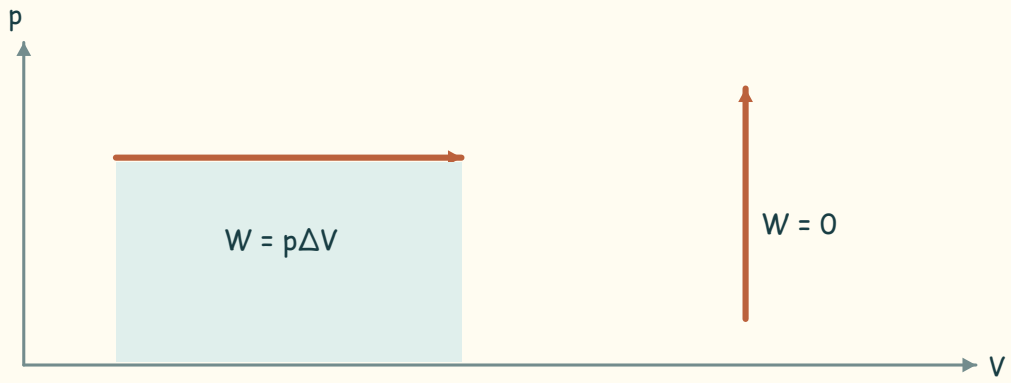


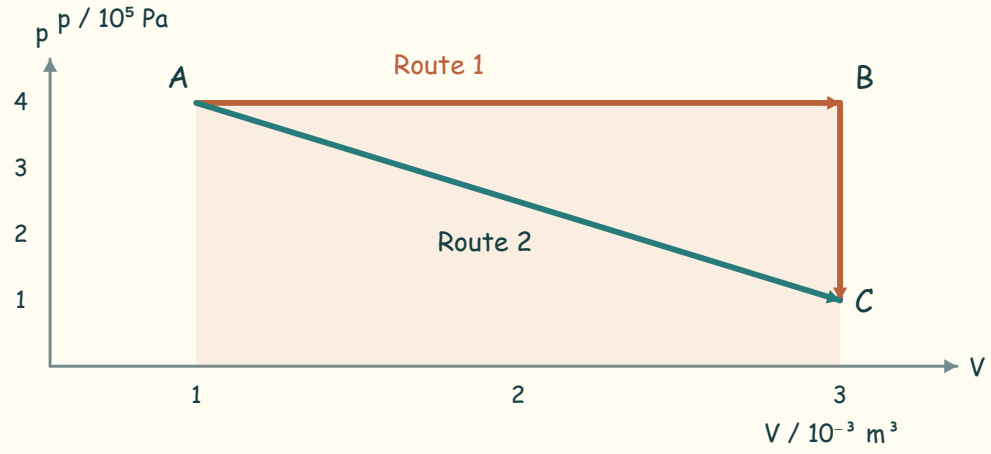
Figure 7. Left: horizontal line (constant pressure) with expansion and compression arrows. Right: vertical line (constant volume), "area $= 0$, $W = 0$".

Constant pressure is a horizontal line. In an expansion, work is done by the gas on the surroundings, so less of the heat stays as internal energy: $\Delta U = Q - W$. In a compression, work is done on the gas, which raises its internal energy. Squeeze the gas into a smaller volume and the molecules hit the walls more often — higher pressure. Constant volume is a vertical line. The area under it is zero, so no work is done. Work is only done when the volume changes.

WORKED PROBLEM 03 / 13

Problem 3 — Two Paths: U, Q and W

Tsokos (CUP) B.4 · First law, state functions · IB Paper 2 style extension



Visual setup · Two Paths: U, Q and W

Return to the gas in Problem 2.

(a) Using $U = (\frac{3}{2})pV$, calculate the change in internal energy from A to C.

(b) Calculate the heat supplied along Route 1 and along Route 2.

(c) Explain what this tells you about U, Q and W.

▼ Worked solution

(a) $U_A = 1.5 \times (4.0 \times 10^5)(1.0 \times 10^{-3}) = 600\text{J}$

$U_C = 1.5 \times (1.0 \times 10^5)(3.0 \times 10^{-3}) = 450\text{J}$

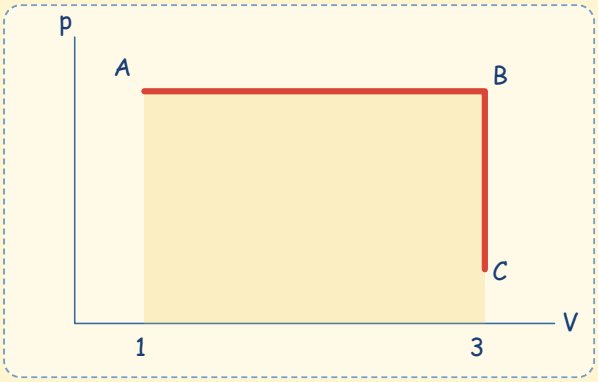
$\Delta U = -150\text{J}$ (the same for both routes)

(b) $Q = \Delta U + W$ Route 1: $Q = -150 + 800 = +650\text{J}$ Route 2: $Q = -150 + 500 = +350\text{J}$

(c) ΔU depends only on the start and end states — it is a state function. Q and W depend on the path taken. Heat and work are ways of transferring energy, not something a gas contains.

INTERACTIVE ARTIFACT 02 · PATH & AREA

Same endpoints. Different work.



Choose the path

W = +800 J
ΔU = −150 J
Q = +650 J

These values use Problems 2 and 3. The shaded area is work; the sign follows the direction of travel.

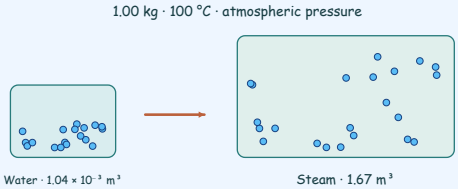
WORKED PROBLEM 04 / 13

Problem 4 — Energy Split During Boiling

Cambridge IB Physics B.4 · First law with change of state · replaces SME: vaporisation of 1.00 mol liquid



QUESTION DATA & MODEL



Visual setup · Energy Split During Boiling

AI-generated notebook illustration · diagram geometry and question data define the physics.

1.00 kg of water at 100 °C is boiled at atmospheric pressure, 1.01×10^5 Pa. The water occupies $1.04 \times 10^{-3}\ m^3$, and the steam produced occupies $1.67 m^3$. The specific latent heat of vaporisation is $2.26 \times 10^6\ \text{J kg}^{-1}$.

(a) Calculate the work done by the system as it expands against the atmosphere.

(b) Determine the increase in internal energy.

(c) What percentage of the latent heat is "spent" pushing the atmosphere back? Where does the rest go?

▼ Worked solution

(a) $\Delta V = 1.67 - 0.00104 \approx 1.669 m^3$

$W = p\Delta V = (1.01 \times 10^5)(1.669) = 1.69 \times 10^5\ \text{J}$

(b) $Q = mL = 2.26 \times 10^6\ \text{J}$

$\Delta U = Q - W = 2.26 \times 10^6 - 1.69 \times 10^5 = 2.09 \times 10^6\ \text{J}$

(c) $\frac{1.69 \times 10^5}{2.26 \times 10^6} \approx 7.5\%$. The remaining ~92.5% increases the internal energy. The temperature does not change during boiling, so this energy is not kinetic — it goes into potential energy, pulling the molecules apart against their intermolecular forces. (So steam at 100 °C is not an ideal gas situation during the change of state.) The First Law says energy is conserved, but not which direction a process goes. Heat flows from hot to cold, never the reverse by itself. To describe this direction we need entropy.

04 4. ENTROPY

FIGURE 08

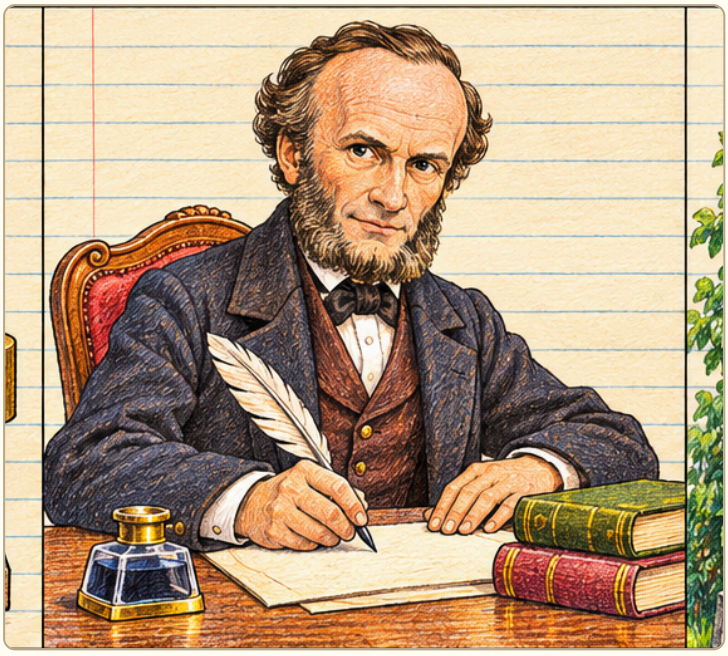


Figure 8. Rudolf Clausius, 1865, writing a new Greek-derived word: "entropy" — from *tropē*, "transformation".

AI-generated notebook illustration · diagram geometry and question data define the physics.

In 1865, Rudolf Clausius named a quantity in his heat equations that only increases in real processes. He took it from the Greek word for "transformation": entropy. The entropy of a system is a measure of the number of possible arrangements of its particles and their energies. In plain words: how disordered the system is. More disorder, more entropy.

FIGURE 09

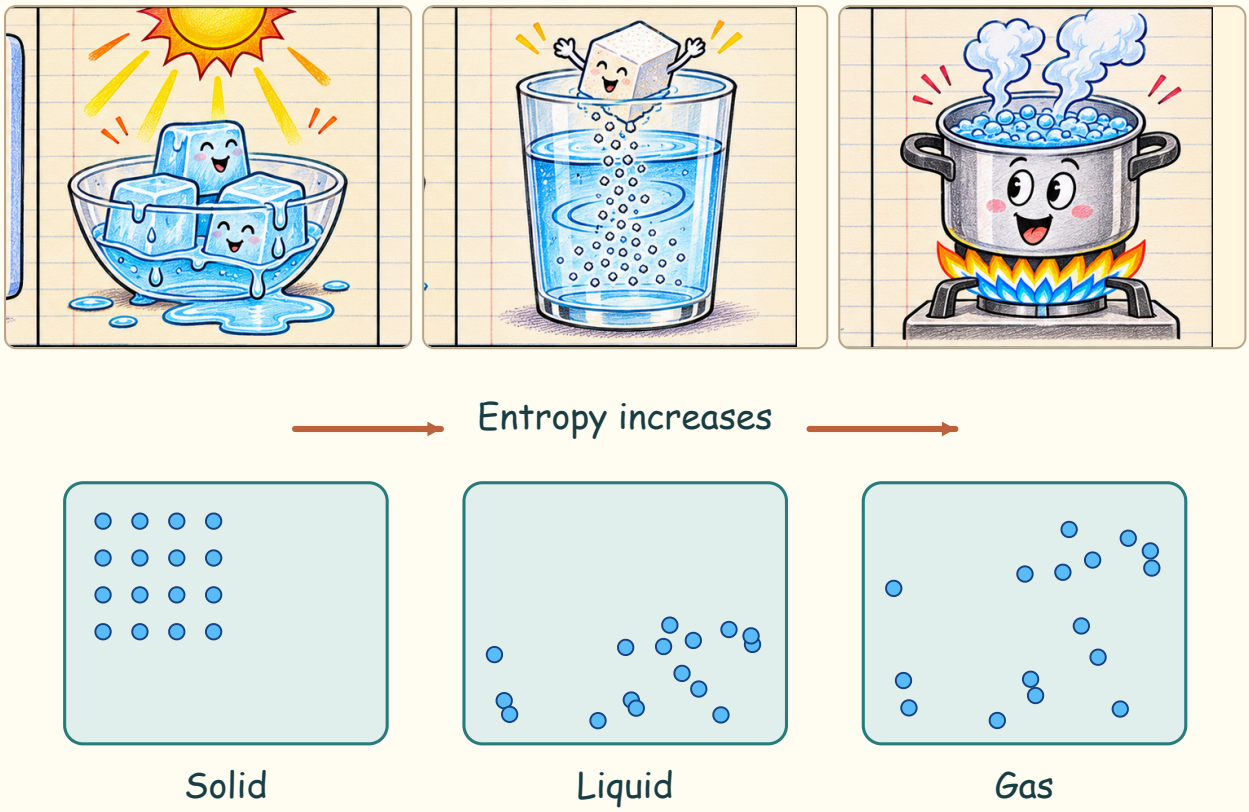


Figure 9. Three boxes — solid (neat lattice), liquid (loosely packed), gas (particles spread out, random).

AI-generated notebook illustration · diagram geometry and question data define the physics.

The ranking of states, from most disordered to least:

gas > liquid > solid

Entropy increases when a substance melts (solid → liquid) or boils (liquid → gas). Raising temperature makes the particles vibrate more; in a gas they move freely and are far apart. Entropy decreases when a substance condenses (gas → liquid) or freezes (liquid → solid). Particles come together, arrange more regularly, and can move less. Fewer ways to arrange the energy means lower entropy.

FIGURE 10



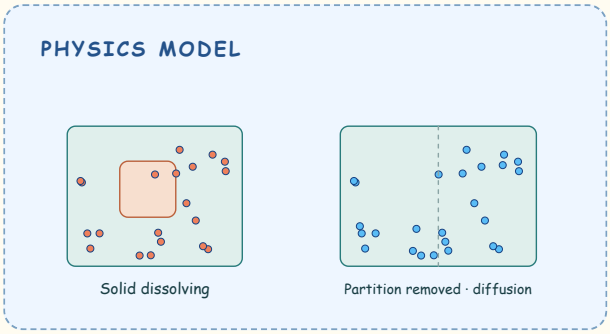


Figure 10. Left: a sugar cube dissolving into a glass of water. Right: a box with a partition, gas on one side; partition removed, gas fills the box.

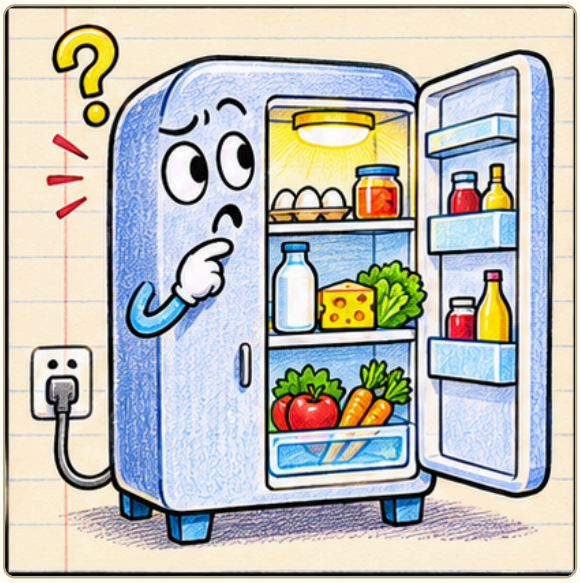
AI-generated notebook illustration · diagram geometry and question data define the physics.

Entropy also increases when: a solid dissolves in a solvent, and a gas diffuses through a container. In both cases, the particles spread out, and there are more ways of arranging the energy.

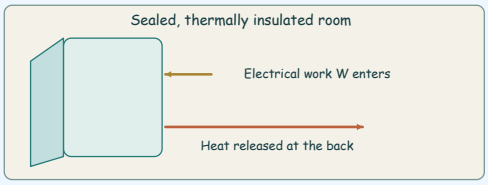
WORKED PROBLEM 05 / 13

Problem 5 — Fridge Door Open in a Sealed Room

Tsokos (CUP) B.4 · Second law / heat pumps · replaces SME: freezer MCQ



QUESTION DATA & MODEL



Visual setup · Fridge Door Open in a Sealed Room

AI-generated notebook illustration · diagram geometry and question data define the physics.

A refrigerator is switched on and running, with its door wide open, in a sealed and thermally insulated room.

(a) Over time, the temperature of the room A. decreases B. increases C. stays the same D. decreases and then returns to its original value

(b) Explain your answer using the First Law, and comment on the total entropy of the room.

▼ Worked solution

(a) B — increases.

(b) The fridge is a heat pump. It takes heat Q_C from its inside and dumps heat Q_H out of the back, where $Q_H = Q_C + W$, and W is the electrical work done by the compressor. With the door open, the "cold" air just mixes back into the room, so the Q_C cancels out. The net effect is that

the electrical energy W is converted into internal energy of the room, every second. The room warms up. The contents of the fridge become colder (less entropy), but the room gains more entropy than that, so the total entropy increases.

Real isolated systems

FIGURE 11



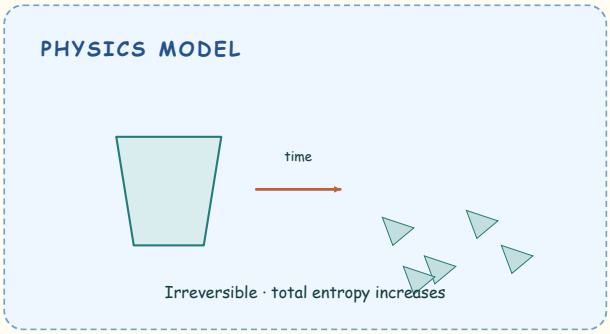


Figure 11. A video clip played forwards (a glass shatters) and backwards (shards jump back into a glass) — "which one looks fake?"

AI-generated notebook illustration · diagram geometry and question data define the physics.

At the macroscopic level, the difference between reversible and irreversible processes is everything. Reversible process: a process where there is no overall change in entropy, as the system and its surroundings are returned to their original states. Irreversible process: a process that results in an increase in entropy, as the system and its surroundings cannot return to their original states. Processes in real isolated systems are almost always irreversible. So the entropy of a real isolated system always increases.

Non-isolated systems — local order, at a price

An isolated system is one where neither matter nor energy can be transferred in or out. In a non-isolated system, matter and energy can move in and out. You can think of it as part of a bigger isolated system. So the entropy of a non-isolated system can decrease locally (e.g. a freezer making ice, a plant making sugar) but only if the surroundings gain an equal or greater amount of entropy.

05 5. CALCULATING CHANGES IN ENTROPY

Macroscopic entropy At a constant temperature T:

$$\Delta S = \frac{\Delta Q}{T}$$

ΔS = change in entropy (JK^{-1})

ΔQ = heat given to or removed from the system (J)

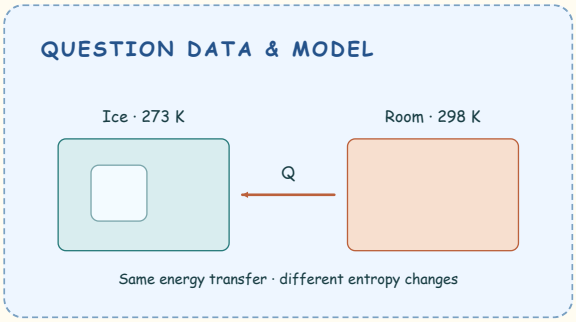
T = temperature of the system (K) Heat given to a system: $\Delta Q > 0$, so $\Delta S > 0$ — entropy increases. Heat removed from a system: $\Delta Q < 0$, so $\Delta S < 0$ — entropy decreases. For a reversible process that returns the system to its original state, the overall entropy change of the system is zero, because entropy depends only on the state of the system. Because T is in the denominator, the same heat causes a bigger entropy change in a cold body than in a hot one.

WORKED PROBLEM 06 / 13

Problem 6 — Ice Melting in a Warm Room

Tsokos (CUP) B.4 · $\Delta S = \frac{\Delta Q}{T}$ · new problem (no SME equivalent)





Visual setup · Ice Melting in a Warm Room

AI-generated notebook illustration · diagram geometry and question data define the physics.

0.50 kg of ice at 0 °C is placed in a large room at 25 °C and melts completely. The room's temperature stays effectively constant. Specific latent heat of fusion of ice $= 3.34 \times 10^5\ \text{J kg}^{-1}$.

(a) Calculate the entropy change of the ice.

(b) Calculate the entropy change of the room.

(c) Calculate the total entropy change, and explain what it tells you.

▼ Worked solution

$Q = mL = 0.50 \times 3.34 \times 10^5 = 1.67 \times 10^5\ \text{J}$

(a) $\Delta S_{\text{ice}} = \frac{+1.67 \times 10^5}{273} = +612\ \text{J K}^{-1}$

(b) $\Delta S_{\text{room}} = \frac{-1.67 \times 10^5}{298} = -560\ \text{J K}^{-1}$

(c) $\Delta S_{total} = 612 - 560 = +52 JK^{-1}$ The ice gains more entropy than the room loses, because it is colder. Total entropy increases, so the process is spontaneous. The reverse (water freezing by itself in a warm room) would decrease total entropy, so it does not happen.

Microscopic entropy

FIGURE 12

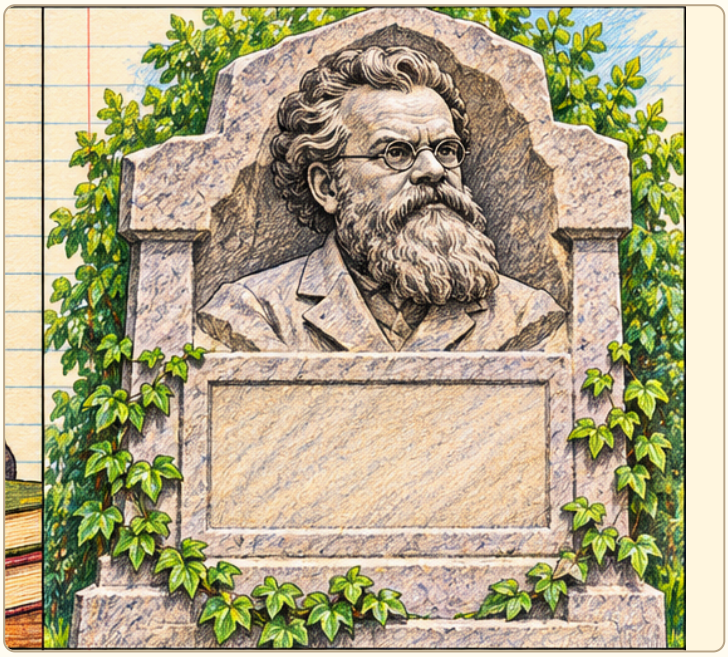


Figure 12. Ludwig Boltzmann's grave in the Zentralfriedhof, Vienna, with the equation $S = k \log W$ carved above his bust.

AI-generated notebook illustration · diagram geometry and question data define the physics.

Ludwig Boltzmann linked entropy to the number of ways particles can be arranged, at a time when many scientists doubted atoms existed. His equation is carved on his grave in Vienna:

$$S = k_B \ln \Omega$$

S = entropy of a system of microscopic particles (JK^{-1})

k_B = Boltzmann constant

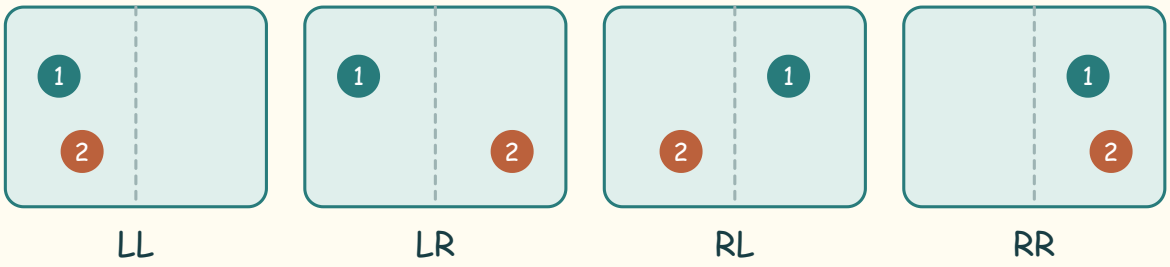
Ω = number of possible microstates of the system And when the number of microstates changes from Ω_1 to Ω_2 :

$$\Delta S = k_B \ln\left(\frac{\Omega_2}{\Omega_1}\right)$$

A microstate is one possible arrangement of the particles in the system. A state can be described by any microscopic or macroscopic property — positions or velocities of molecules, energy, volume, and so on.

FIGURE 13

Two labelled particles · four arrangements

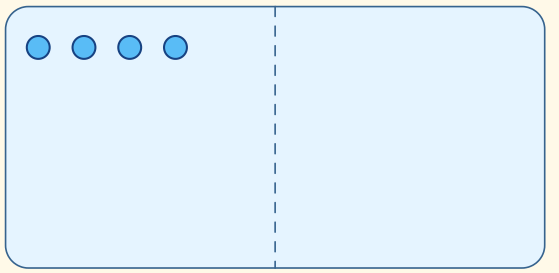


$$\Omega_2 = 2^2 = 4$$

Figure 13. A two-compartment box. Top: all N labelled particles locked in the left compartment. Bottom: partition removed; for $N = 2$, the four arrangements (LL, LR, RL, RR) shown.

INTERACTIVE ARTIFACT 03 · MICROSTATES

Count the possibilities.



One highlighted arrangement: every labelled particle on the left.

Number of distinguishable particles

N = 4

Ω = 16

P(all left) = 1 / 16 = 0.06250

Each particle has two compartment choices. Total arrangements: 2^N.

Take a container with two compartments and N distinguishable particles (each particle can be individually identified).

Initially, all N are sealed in one compartment. We know exactly where every particle is. Only one arrangement is possible: $\Omega_1 = 1^N = 1$. (For $N = 2$ or $N = 4$: $1^2 = 1^4 = 1$.) Remove the partition. Each particle can now be in either compartment, so $\Omega_2 = 2^N$. For $N = 2$, there are $2^2 = 4$ arrangements. For $N = 4$, there are $2^4 = 16$. The entropy change:

$$\Delta S = k_B \ln\left(\frac{2^N}{1^N}\right) = k_B \ln(2^N)$$

Now see it macroscopically. The gas's volume goes from V to 2V. Its entropy gain is:

$$\Delta S = Nk_B \ln(2V) - Nk_B \ln(V) = Nk_B \ln\left(\frac{2V}{V}\right) = Nk_B \ln 2$$

This is the same as $k_B \ln(2^N)$: counting arrangements and using volume give the same result.

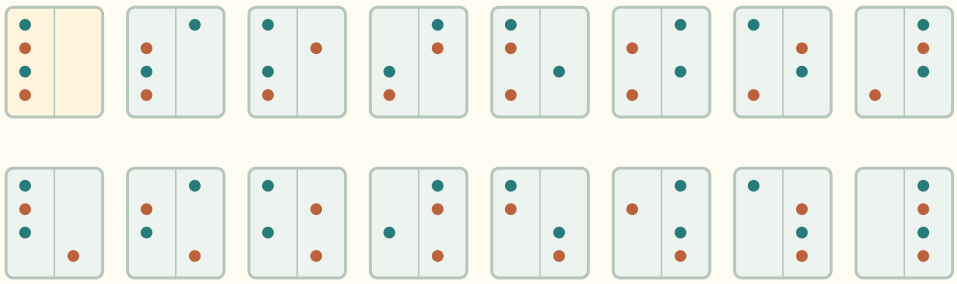
Examiner tip: Entropy underpins ideas from quantum mechanics to the Schwarzschild radius of a black hole, so it's fine if it feels strange at first. Think of microstates as a measure of how certain we are about the system. We're more certain where atoms are in a solid than in a gas, so the solid has lower entropy. Raise the temperature, pressure or volume of a gas and we become less certain where the atoms are, so entropy goes up.

WORKED PROBLEM 07 / 13

Problem 7 — Microstates and Probability

Tsokos (CUP) B.4 · $S = k\ln\Omega$, microstates · new problem

Four labelled molecules · 16 possible arrangements



Highlighted: all four in the left half

Visual setup · Microstates and Probability

(a) Four distinguishable gas molecules are in a box divided into two equal halves. Calculate the probability that all four are found in the left half at a given instant.

(b) Repeat for 100 molecules.

(c) One mole of an ideal gas expands freely into a vacuum, doubling its volume. Calculate its change in entropy.

(d) Explain why, although the laws of motion of each molecule are reversible, the gas never spontaneously goes back into half the box.

▼ Worked solution

(a) Total microstates $= 2^4 = 16$. Only 1 has all four on the left. $P = \frac{1}{16} = 0.0625$

(b) $P = \frac{1}{2^{100}} \approx 7.9 \times 10^{-31}$

(c) $\Delta S = Nk_B \ln 2 = nR\ln 2 = 1 \times 8.31 \times 0.693 = 5.76\,\text{JK}^{-1}$

(d) A real gas has about 10^{23} molecules. "All in one half" is one arrangement out of $2^{10^{23}}$. It isn't forbidden, it is just so improbable that it would not happen even once in the age of the universe. The Second Law is a law of overwhelming probability.

06 6. THE SECOND LAW OF THERMODYNAMICS

FIGURE 14

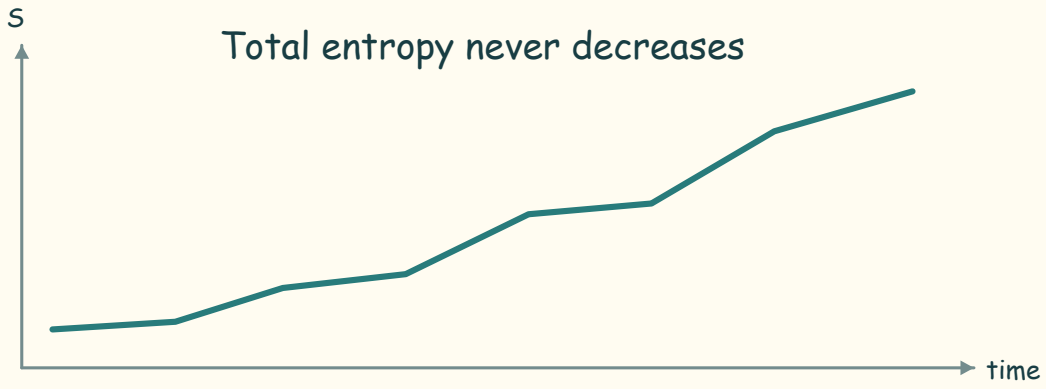


Figure 14. An arrow labelled "time" pointing to the right, with an entropy graph climbing alongside it.

The Second Law of Thermodynamics:

In every process, the total entropy of an isolated system always increases. For a non-isolated system, we widen the boundary to include everything: the entropy of the Universe must increase.

There are two more practical forms of this law. The Clausius form: Thermal energy cannot spontaneously transfer from a region of lower temperature to a region of higher temperature. The key word is spontaneously. A heat pump — like your fridge or AC — does move heat from cold to hot, but only because work is done on it. The Kelvin form: When extracting energy from a heat reservoir, it is impossible to convert it all into work.

FIGURE 15

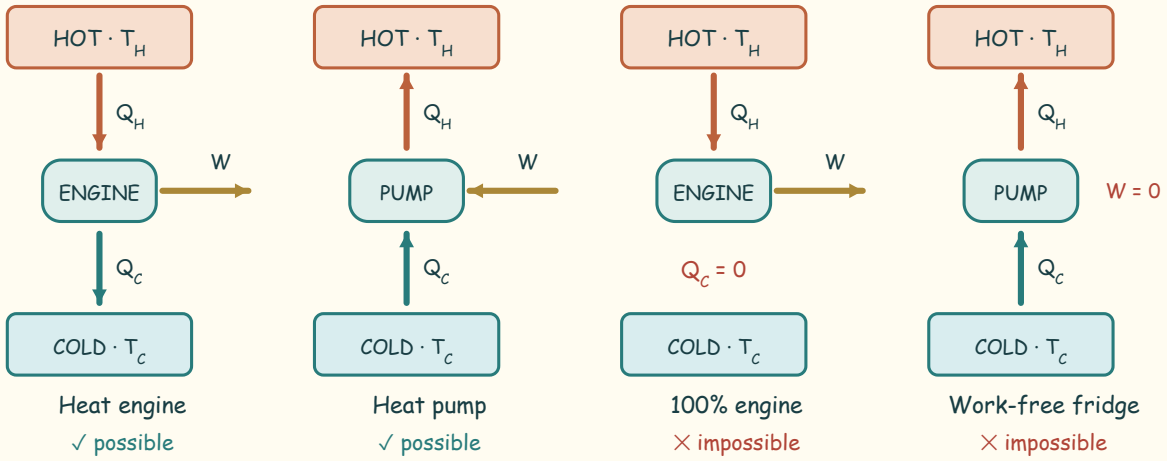


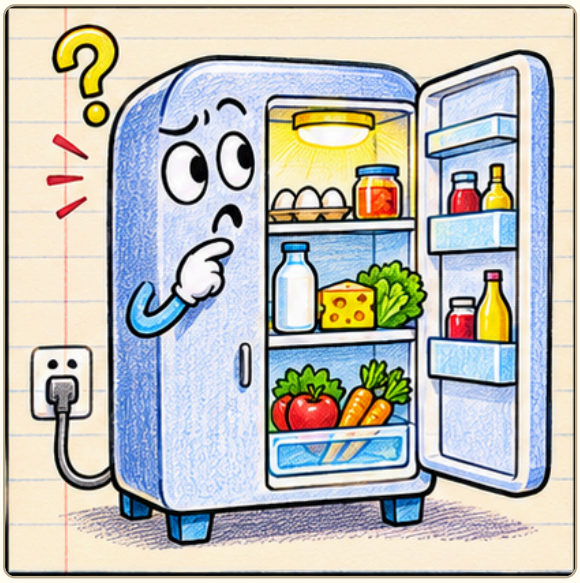
Figure 15. Four engine/pump diagrams with hot reservoir at top and cold reservoir below. (1) Normal heat engine: Q_H in, W out, Q_C out. (2) Normal heat pump: W in, Q_C taken from cold, Q_H delivered to hot. (3) ✗ "Perfect engine": Q_H in, all converted to W, nothing to cold — violates Kelvin. (4) ✗ "Perfect fridge": heat moves cold → hot with no work — violates Clausius.

Machines (3) and (4) cannot exist: they violate the Second Law.

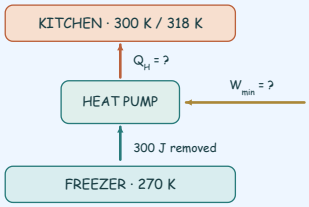
WORKED PROBLEM 08 / 13

Problem 8 — Minimum Work for a Freezer

Cambridge IB Physics B.4 · Clausius statement & refrigerators · new problem



QUESTION DATA & MODEL



Visual setup · Minimum Work for a Freezer

AI-generated notebook illustration · diagram geometry and question data define the physics.

A freezer removes 300 J of heat from its inside, at 270 K, and releases heat into a kitchen at 300 K.

(a) Using the condition that the total entropy cannot decrease, determine the minimum heat that must be released into the kitchen.

(b) Hence find the minimum work the compressor must do.

(c) Explain why a freezer in Lucknow in May (kitchen at 318 K) needs more work for the same job.

▼ Worked solution

(a) Entropy lost by the inside: $\frac{300}{270} = 1.11\,\text{JK}^{-1}$. The kitchen must gain at least that: $\frac{Q_H}{300} \geq \frac{300}{270} \rightarrow Q_H \geq \frac{300 \times 300}{270} = 333\,\text{J}$

(b) $W = Q_H - Q_C = 333 - 300 = 33\,\text{J}$ (minimum)

(c) $Q_{H,min} = \frac{300 \times 318}{270} = 353\text{J} \rightarrow W_{min} = 53\text{J}$. A hotter kitchen means each joule dumped there carries less entropy, so you must dump more joules — and that extra comes from work.

07 7. THERMODYNAMIC PROCESSES

The four main thermodynamic processes (all for an ideal gas):

Process	What's constant	Key result
Isovolumetric	Volume	$W = 0$
Isobaric	Pressure	$\Delta p = 0$
Isothermal	Temperature	$\Delta T = 0$
Adiabatic	No heat transfer	$Q = 0$

Constant pressure (isobaric)

FIGURE 16

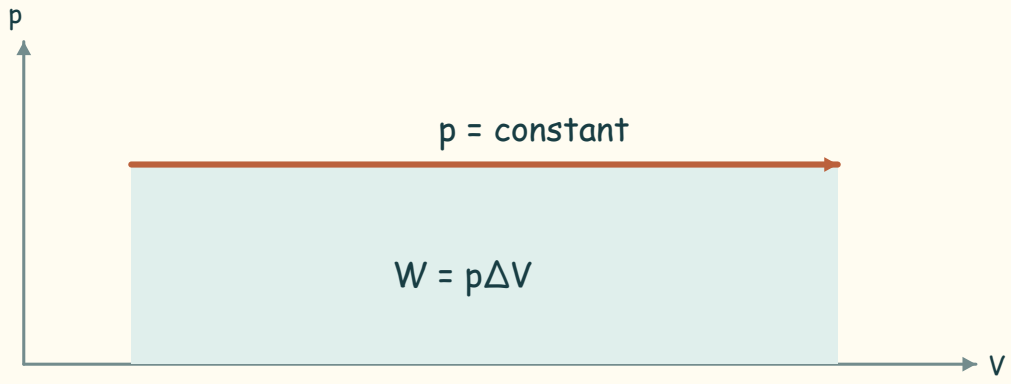


Figure 16. p–V diagram, horizontal line with an arrow.

An isobaric process is one in which no change in pressure occurs. It happens when a gas is allowed to expand or contract freely as its temperature changes. $W = p\Delta VQ = \Delta U + WQ = \Delta U \pm p\Delta V$

The ± shows whether work is done by or on the gas.

Constant volume (isovolumetric)

FIGURE 17

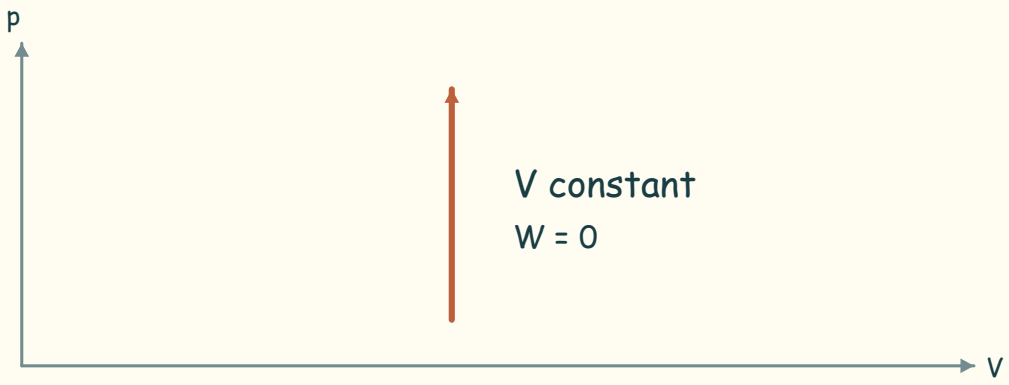


Figure 17. p–V diagram, vertical line with an arrow.

An isovolumetric process is one where no change in volume occurs, so the system does no work. $W = 0$:

$$Q = \Delta U + 0 \Rightarrow Q = \Delta U$$

Constant temperature (isothermal)

FIGURE 18

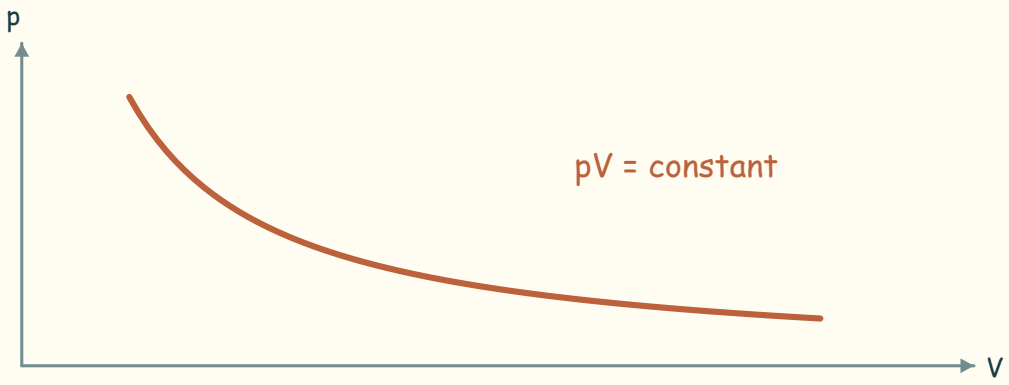


Figure 18. p–V diagram, a smooth curve (pV = constant).

An isothermal process is one in which no change in temperature occurs. No temperature change means no change in internal energy, $\Delta U = 0$:

$$Q = 0 + W \Rightarrow Q = W$$

No heat transfer (adiabatic)

FIGURE 19

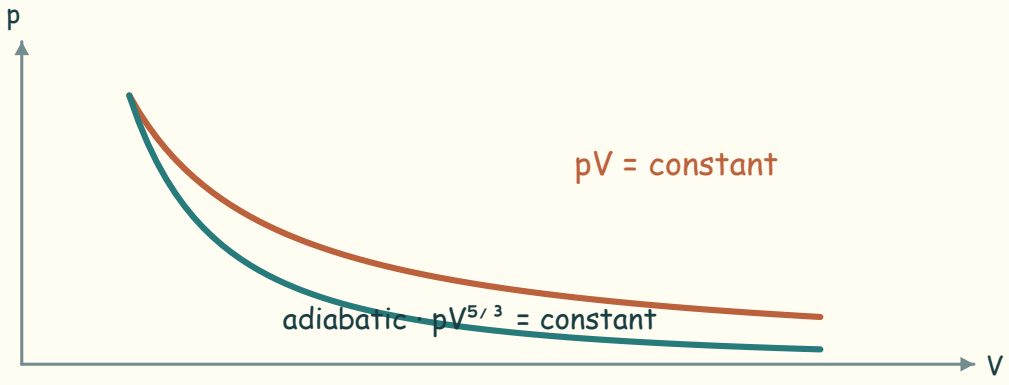


Figure 19. p–V diagram showing an adiabatic curve steeper than the isothermal curve through the same point.

An adiabatic process is one where no heat is transferred into or out of the system. $Q = 0$:

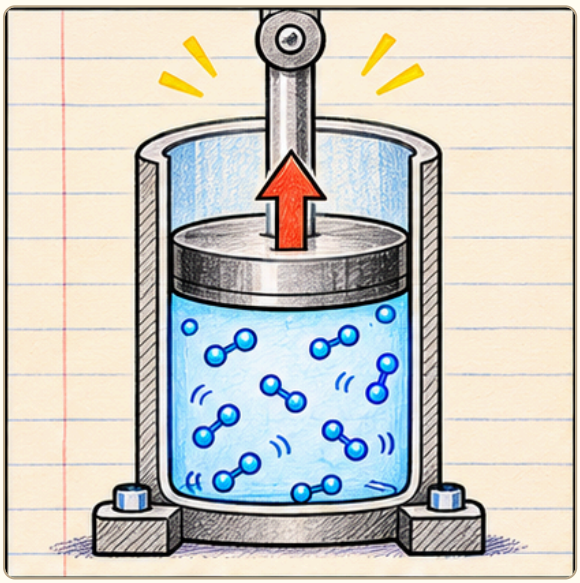
$0 = \Delta U + W \Rightarrow W = -\Delta U$

All the work is done at the expense of the gas's internal energy. So an adiabatic process usually comes with a temperature change. That's why the adiabatic curve is steeper than the isothermal.

WORKED PROBLEM 09 / 13

Problem 9 — Same Heat, Three Processes

Tsokos (CUP) B.4 · Isobaric / isovolumetric / isothermal · replaces SME: gases X, Y, Z table



QUESTION DATA & MODEL

Same input: 500 J of heat

Constant volume	Constant pressure	Constant temperature
W = ?	W = ?	W = ?
ΔU = ? ΔT = ?	ΔU = ? ΔT = ?	ΔU = ? ΔT = ?

0.20 mol · ideal monatomic gas

Visual setup · Same Heat, Three Processes

AI-generated notebook illustration · diagram geometry and question data define the physics.

500 J of heat is supplied to 0.20 mol of an ideal monatomic gas. Compare what happens if the heating is (i) isovolumetric, (ii) isobaric, (iii) isothermal.

(a) For each, find W, ΔU and ΔT.

(b) Explain why the isobaric gas warms up less.

▼ Worked solution

(i) Isovolumetric: $W = 0$, so $\Delta U = Q = 500\text{J}$.

$$\Delta T = \frac{\Delta U}{(1.5nR)} = \frac{500}{(1.5 \times 0.20 \times 8.31)} = +201\text{K}$$

(ii) Isobaric: $W = p\Delta V = nR\Delta T$ and $\Delta U = 1.5nR\Delta T$, so $Q = 2.5nR\Delta T$.

$\Delta T = \frac{500}{(2.5 \times 0.20 \times 8.31)} = +120\text{K}$

$\Delta U = 1.5 \times 0.20 \times 8.31 \times 120 = +300\text{J}$, $W = +200\text{J}$

(iii) Isothermal: $\Delta T = 0$, $\Delta U = 0$, $W = Q = +500\text{J}$

(b) At constant pressure the gas has to expand, so 40% of the heat leaves as work pushing on the surroundings. Only 60% stays to raise the temperature. (This is why a gas has a bigger "specific heat" at constant pressure than at constant volume.)

Process	W	ΔU	ΔT
Isovolumetric	0	+500 J	+201 K
Isobaric	+200 J	+300 J	+120 K
Isothermal	+500 J	0	0

Entropy in thermodynamic processes

Recall: at constant temperature, $\Delta S = \frac{\Delta Q}{T}$. Heat gained → entropy increases. Heat lost → entropy decreases. A reversible process that returns the system to its original state → $\Delta S = 0$.

Process	Change	Heat gained or lost, ΔQ	Change in entropy, ΔS
Isothermal	Expansion	$\Delta Q > 0$: heat gained = work done by gas	$\Delta S > 0$, increases
Isothermal	Compression	$\Delta Q < 0$: heat lost = work done on gas	$\Delta S < 0$, decreases
Isobaric	Expansion	$\Delta Q > 0$: heat gained = increase in U + work done by gas	$\Delta S > 0$, increases
Isobaric	Compression	$\Delta Q < 0$: heat lost = decrease in U + work done on gas	$\Delta S < 0$, decreases

Process	Change	Heat gained or lost, ΔQ	Change in entropy, ΔS
Isovolumetric	Pressure rise	$\Delta Q > 0$: heat gained, temperature rises	$\Delta S > 0$, increases
Isovolumetric	Pressure drop	$\Delta Q < 0$: heat lost, temperature drops	$\Delta S < 0$, decreases
Adiabatic	Expansion	$\Delta Q = 0$: p and T decrease, no heat gained/lost	$\Delta S = 0$, no change
Adiabatic	Compression	$\Delta Q = 0$: p and T increase, no heat gained/lost	$\Delta S = 0$, no change

Adiabatic processes

FIGURE 20

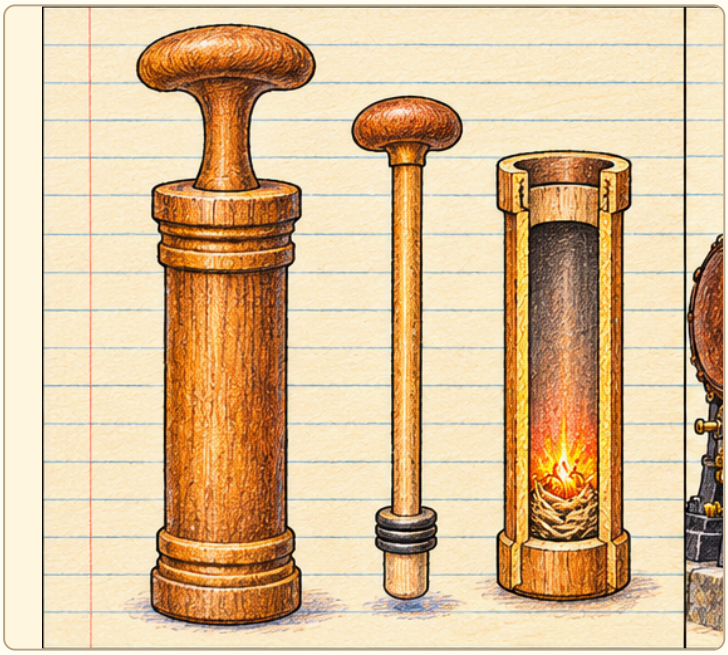


Figure 20. A traditional fire piston: a sealed tube with a plunger; a tiny piece of tinder at the bottom glowing after the plunger is slammed down.

AI-generated notebook illustration · diagram geometry and question data define the physics.

The fire piston, used traditionally in Southeast Asia, is a tube with a tight plunger: pushing it down quickly heats the air enough to ignite tinder. In 1892 Rudolf Diesel used the same idea in his engine, where compressed air ignites the fuel without a spark plug. Both work because rapid compression is adiabatic: the work done on the gas raises its temperature. For an ideal monatomic gas:

$$pV^{\frac{5}{3}} = \text{constant} \cdot p_1 V_1^{\frac{5}{3}} = p_2 V_2^{\frac{5}{3}}$$

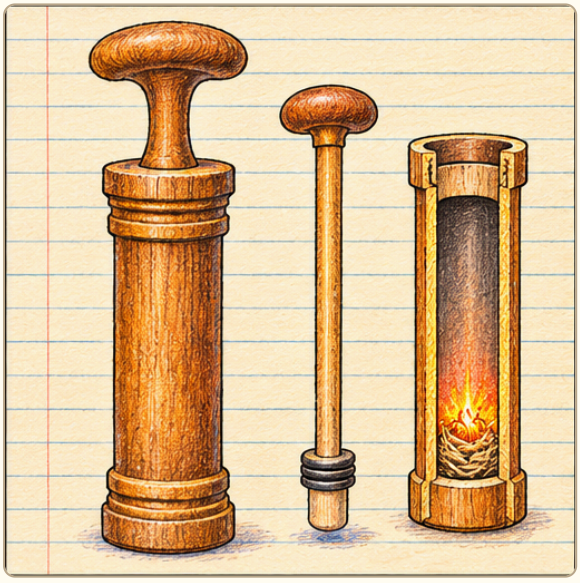
p_1, p_2 = initial and final pressure (Pa) V_1, V_2 = initial and final volume (m^3) Combining with $pV = nRT$ (substitute $p = \frac{nRT}{V}$) gives the temperature form:

$$T_1 V_1^{\frac{2}{3}} = T_2 V_2^{\frac{2}{3}}$$

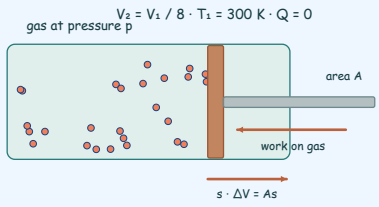
WORKED PROBLEM 10 / 13

Problem 10 — Adiabatic Compression in a Fire Piston

Tsokos (CUP) B.4 · Adiabatic $pV^{\frac{5}{3}}$ = const · replaces SME: adiabatic p and T calculations



QUESTION DATA & MODEL



Visual setup · Adiabatic Compression in a Fire Piston

AI-generated notebook illustration · diagram geometry and question data define the physics.

Air in a fire piston is modelled as an ideal monatomic gas. It starts at 300 K, 1.0×10^5 Pa and volume $1.0 \times 10^{-3}\ m^3$ (scaled-up model), and is compressed adiabatically to one- eighth of its volume.

(a) Calculate the final temperature.

(b) Calculate the final pressure.

(c) Calculate the work done on the gas.

(d) The tinder ignites at about 530 K. Find the smallest compression ratio $\frac{V_1}{V_2}$ that would light it.

▼ Worked solution

(a) $T_2 = T_1\left(\frac{V_1}{V_2}\right)^{\frac{2}{3}} = 300 \times 8^{\frac{2}{3}} = 300 \times 4 = 1200\text{K}$

$p_2 = p_1\left(\frac{V_1}{V_2}\right)^{\frac{5}{3}}$

(b) $= 1.0 \times 10^5 \times 8^{\frac{5}{3}}$ (Check: $\frac{pV}{T}$ should be constant: $\frac{100}{300} = \frac{3200 \times \frac{1}{8}}{1200} = 0.333$ ✓)

$= 1.0 \times 10^5 \times 32$

$= 3.2 \times 10^6\,\text{Pa}$

(c) $nR = \frac{p_1 V_1}{T_1} = \frac{100}{300} = 0.333\text{JK}^{-1}$

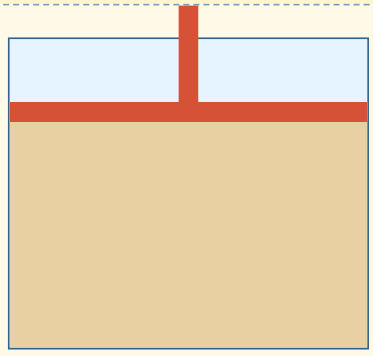
$\Delta U = 1.5 \times 0.333 \times (1200 - 300) = 450\text{J}$

$Q = 0$, so $W = -\Delta U = -450\text{J}$. 450 J of work is done on the gas.

(d) $\left(\frac{V_1}{V_2}\right)^{\frac{2}{3}} = \frac{530}{300} = 1.77 \rightarrow \frac{V_1}{V_2} = 1.77^{1.5} \approx 2.4$ (Real air is diatomic, so the real ratio needed is larger — but the idea holds.)

INTERACTIVE ARTIFACT 04 · FIRE PISTON

Compress the gas. Watch it heat.



Ideal monatomic model from Problem 10; $T_1 = 300$ K.

Compression ratio V_1 / V_2

1.0 : 1

$T_2 = 300$ K

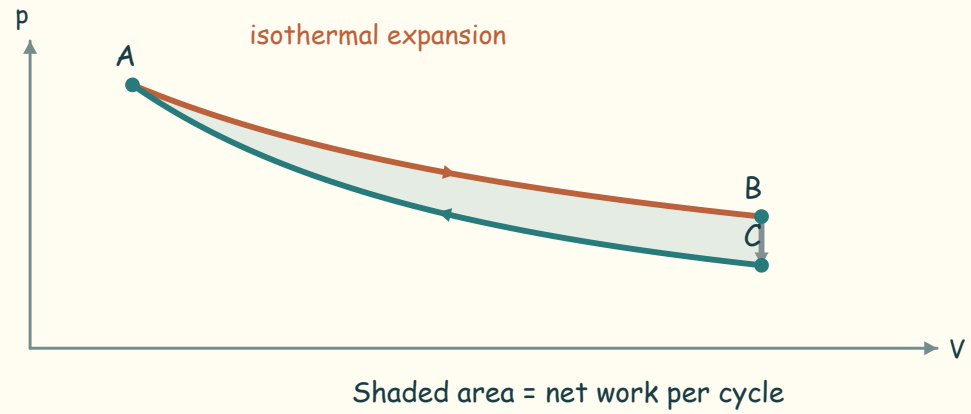
$p_2 = 1.00 \times 10^5$ Pa

Below the stated tinder ignition temperature of 530 K.

WORKED PROBLEM 11 / 13

Problem 11 — Full Cycle with Entropy (HL)

IB HL Paper 2 style · Cycles & entropy · replaces SME: ABCA cycle sketch



Visual setup · Full Cycle with Entropy (HL)

1.0 mol of an ideal monatomic gas runs the cycle $A \to B \to C \to A$: $A \to B$: isothermal expansion at 400 K, from volume V to 2V $B \to C$: isovolumetric change $C \to A$: adiabatic compression back to A For an isothermal change, you are given: $W = nRT\ln(\frac{V_2}{V_1})$.

FIGURE 21

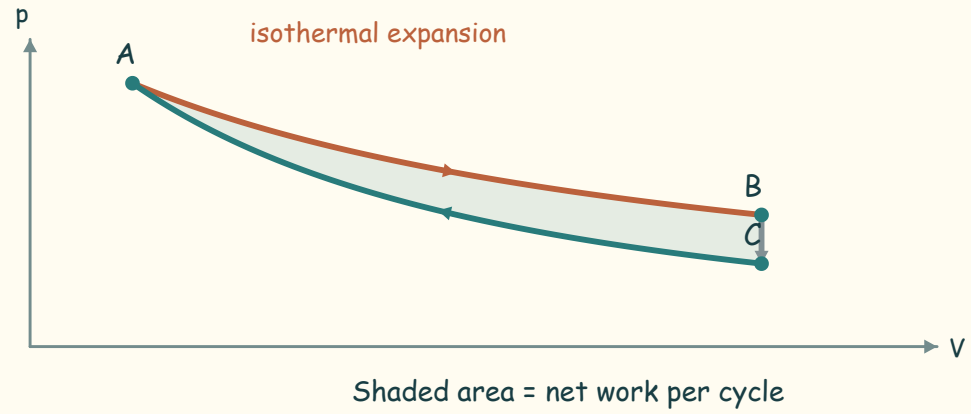


Figure 21. p-V diagram: A (top left) curving down to B (right) along an isotherm; B straight down to C; C curving steeply back up to A along an adiabat.

(a) Sketch the cycle and state whether $B \to C$ is heating or cooling.

(b) Calculate the heat absorbed from $A \to B$ and the entropy change of the gas from $A \to B$.

(c) Calculate the temperature at C.

(d) Calculate the heat released from B → C.

(e) Calculate the net work and efficiency of the cycle.

(f) State where the entropy of the gas is greatest. Compare the efficiency with a Carnot engine between the same extreme temperatures.

▼ Worked solution

(a) C → A is an adiabatic compression, which must end at A. So C must be at a lower pressure than B: B → C is a pressure drop at constant volume — cooling. The line goes straight down.

(b) $Q_{AB} = W = nRT\ln 2 = 1.0 \times 8.31 \times 400 \times 0.693 = 2.30 \times 10^3\ \text{J}$

$\Delta S_{AB} = \frac{Q}{T} = \frac{2304}{400} = +5.76\,\text{JK}^{-1}$ (This equals R ln 2, the same as the free expansion in Problem 7.)

(c) $T_C(2V)^{\frac{2}{3}} = T_A V^{\frac{2}{3}} \rightarrow T_C = 400 \times (\frac{1}{2})^{\frac{2}{3}} = 400 \times 0.630 = 252\text{K}$

(d) $W = 0$, so Q = ΔU = 1.5 nR ΔT = 1.5 × 8.31 × (252 − 400) = −1.84 × 10³ J (1840 J released)

(e) Net work = 2304 − 1845 ≈ 460J

η = 460 / 2304 ≈ 0.20 (20%)

(f) A → B: isothermal expansion, heat in, entropy increases. B → C: cooling, heat out, entropy decreases. C → A: adiabatic, entropy constant. So entropy is greatest at B. Carnot between 400 K and 252 K: $\eta = 1 - \frac{252}{400} = 37\%$. Our cycle reaches only 20%, because heat is rejected over a range of temperatures during B → C rather than at the lowest temperature.

08 8. HEAT ENGINES

FIGURE 22

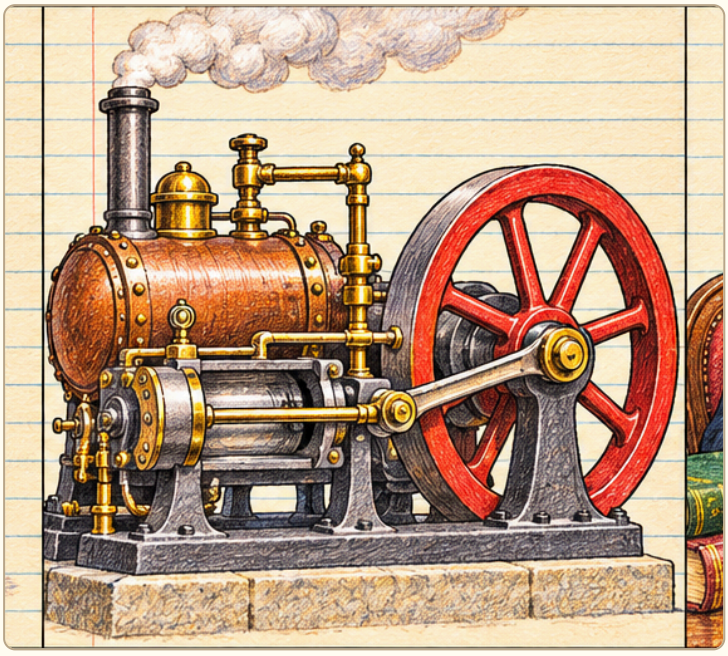
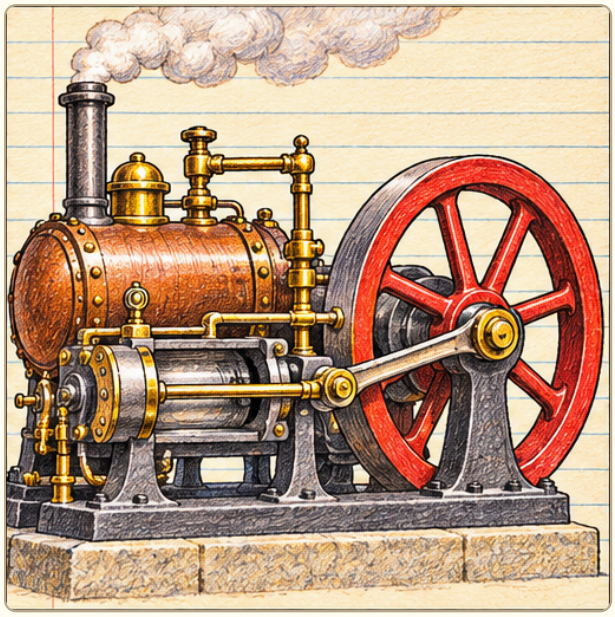


Figure 22. James Watt's steam engine, Industrial Revolution era — pistons, a flywheel, a coal fire under a boiler.

AI-generated notebook illustration · diagram geometry and question data define the physics.

James Watt's improvements to the steam engine powered the Industrial Revolution, and the unit of power, the watt, is named after him. A steam engine is an example of a heat engine. A heat engine is a device that converts thermal energy into mechanical work. It runs through a series of thermodynamic processes that form a closed cycle — a cycle in which the system returns to its initial state.

FIGURE 23



PHYSICS MODEL

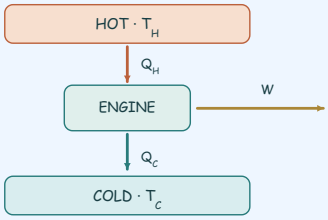


Figure 23. A simple heat engine — gas in a cylinder with a piston, a hot reservoir (T_H) below, a cold reservoir (T_C) to the side.

AI-generated notebook illustration · diagram geometry and question data define the physics.

A simple heat engine is a gas in a cylinder with a piston. Its cycle has four steps:

1. Extract heat from a hot reservoir. A hot reservoir (source of thermal energy) at high temperature T_H transfers heat Q_H into the engine.

2. Use some of that heat to do work. The gas expands and pushes the piston out — mechanical work.

3. Release excess heat into a cold reservoir. The gas cools at constant volume, releasing heat Q_C into a cold reservoir (a sink for excess heat) at lower temperature T_C.

4. Repeat. The piston is pushed back, compressing the gas to its original state. The cycle repeats as many times as needed, continuously converting heat into work.

FIGURE 24

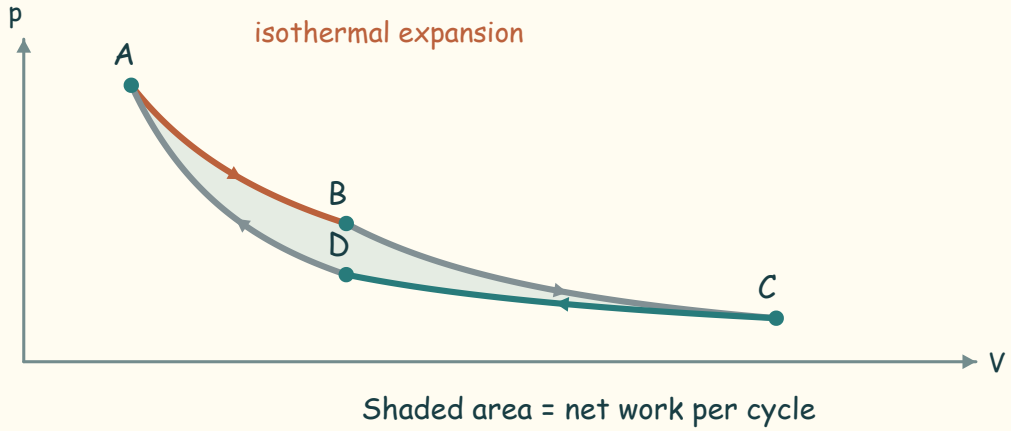


Figure 24. A closed loop on a p–V diagram, interior shaded: "area inside the loop = net work done per cycle". Next to it, the energy-flow diagram: Q_H in from top, W out to the side, Q_C out at the bottom.

For a cyclic process, the p–V diagram is a closed loop, and the area inside the loop is the net work done in one cycle.

$$W_{out} = Q_H - Q_C$$

W_{out} = useful work output of the heat engine (J)

Q_H = heat transferred from hot reservoir to engine (J)

Q_C = heat transferred from engine to cold reservoir (J)

Efficiency of heat engines

The goal: convert heat into useful work as efficiently as possible.

$$\eta = \frac{\text{useful work output}}{\text{total energy input}} = \frac{W_{out}}{Q_H} = \frac{(Q_H - Q_C)}{Q_H} = 1 - \frac{Q_C}{Q_H}$$

η = efficiency

W_{out} = useful work output (J)

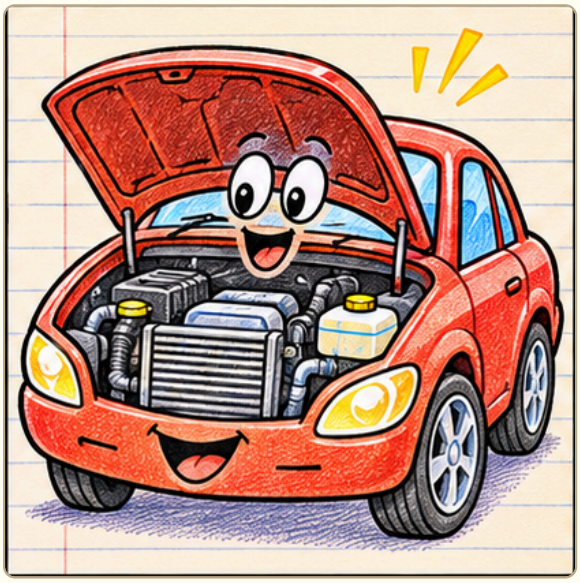
Q_H = total energy input from hot reservoir (J)

Q_C = energy lost to cold reservoir (J)

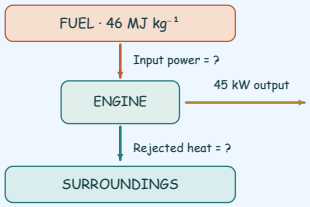
WORKED PROBLEM 12 / 13

Problem 12 — Car Engine Energy Flow

Tsokos (CUP) B.4 · Heat engine efficiency · new problem



QUESTION DATA & MODEL



Visual setup · Car Engine Energy Flow

AI-generated notebook illustration · diagram geometry and question data define the physics.

A car engine has an efficiency of 30% and delivers a mechanical power output of 45 kW. The energy density of petrol is $46 MJkg^{-1}$.

(a) Calculate the rate at which energy is taken from the burning fuel.

(b) Calculate the rate at which heat is rejected to the surroundings.

(c) Calculate the mass of petrol used per second, and per hour.

▼ Worked solution

(a) Q_H per second $\frac{= 45}{0.30} = 150 kW$

(b) Q_C per second $= 150 - 45 = 105\text{kW}$ — more than twice the useful output, lost through the radiator and exhaust.

(c) $\frac{150\times10^3}{46\times10^6} = 3.3\times10^{-3}\,\text{kgs}^{-1} \approx 12\text{kg}$ per hour

09

9. THE CARNOT CYCLE

FIGURE 25

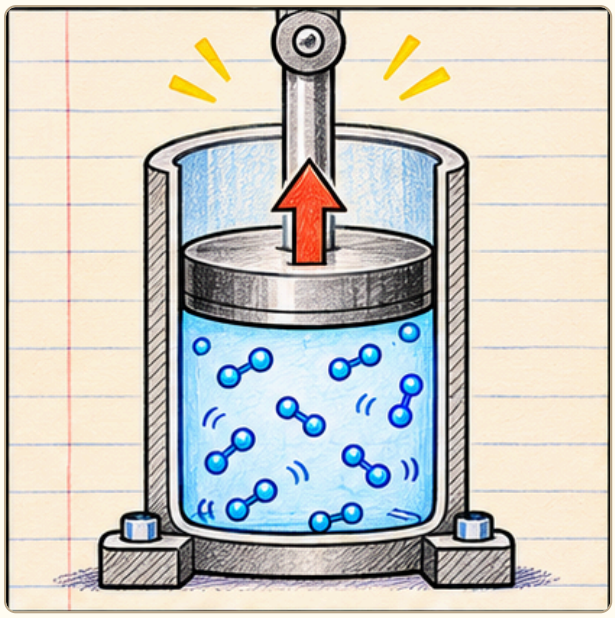


Figure 25. Sadi Carnot, a young French military engineer, 1824, and the cover of his only book, *Réflexions sur la puissance motrice du feu* — "Reflections on the Motive Power of Fire".

AI-generated notebook illustration · diagram geometry and question data define the physics.

In 1824 Sadi Carnot, a French engineer, asked what the maximum possible efficiency of any engine is. His answer, an ideal reversible cycle, was largely ignored until Clapeyron, Kelvin and Clausius built on it. (Carnot died of cholera in 1832, aged 36.) A thermodynamic system running at its greatest possible efficiency follows a cycle called the Carnot cycle.

FIGURE 26



PHYSICS MODEL

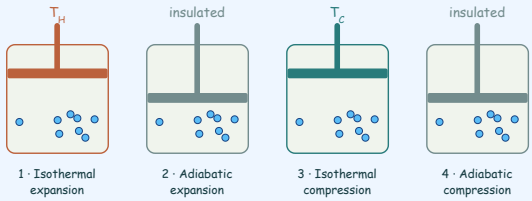


Figure 26. The four stages of the Carnot cycle shown with a gas in a piston: (1) on the hot reservoir, expanding; (2) insulated, expanding; (3) on the cold reservoir, compressed; (4) insulated, compressed.

AI-generated notebook illustration · diagram geometry and question data define the physics.

FIGURE 27

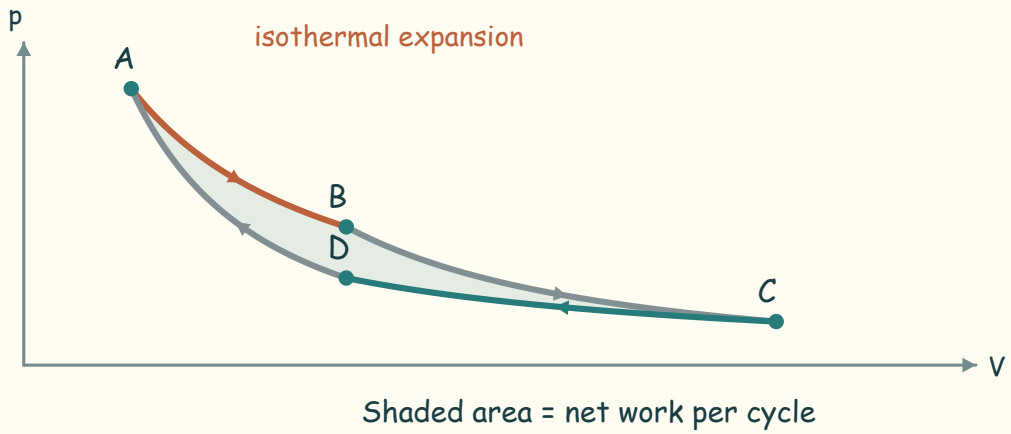


Figure 27. p–V diagram of the Carnot cycle: two isotherms (T_H at top, T_C at bottom) joined by two steep adiabats. Enclosed area shaded = work done per cycle.

The Carnot cycle is an idealised, reversible process with four stages:

1. Isothermal expansion The gas absorbs heat Q_H from the hot reservoir at T_H. It expands (volume increases), doing work, at constant temperature ($\Delta T = 0$). Work done by the gas = heat gained, Q_H.

2. Adiabatic expansion The gas keeps expanding and doing work on the surroundings. Volume increases, pressure decreases. The gas cools from T_H to T_C, with no heat transferred ($\Delta Q = 0$).

3. Isothermal compression The gas is compressed and transfers heat Q_C to the cold reservoir at T_C. Work is done on the gas (volume decreases) at constant temperature. Work done on the gas = heat lost, Q_C.

4. Adiabatic compression The gas is compressed further. Volume decreases, pressure increases. The gas heats up from T_C back to T_H, with no heat transferred ($\Delta Q = 0$). At the end of stage 4, the gas is back to its original state, and the cycle repeats.

FIGURE 28

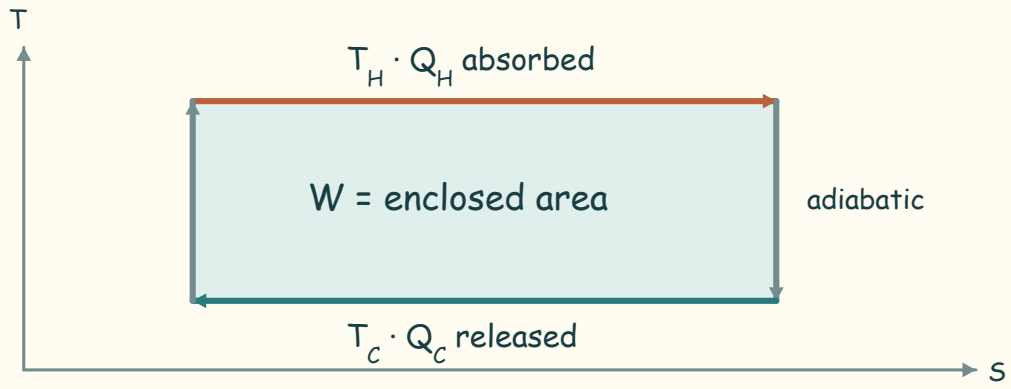


Figure 28. Temperature–entropy (T–S) diagram of the Carnot cycle — a perfect rectangle: horizontal at T_H (entropy rising), vertical drop (adiabatic), horizontal at T_C (entropy falling), vertical rise back.

On a temperature–entropy diagram, the Carnot cycle is a perfect rectangle. The bigger the gap between the hot and cold reservoir temperatures, the higher the efficiency. The maximum theoretical efficiency (Carnot cycle only):

$$\eta_C = 1 - \frac{T_C}{T_H}$$

η_C = maximum theoretical (Carnot) efficiency

T_C = temperature of the cold reservoir (K)

T_H = temperature of the hot reservoir (K)

Deriving Carnot's limit from entropy

This can be derived using entropy.

Step 1 — Isothermal expansion ($A \rightarrow B$). The gas absorbs Q_H at T_H:

$$\Delta S_{AB} = \frac{+Q_H}{T_H}$$

Step 2 — Isothermal compression ($C \rightarrow D$). The gas releases Q_C at T_C:

$$\Delta S_{CD} = \frac{-Q_C}{T_C}$$

Step 3 — Net entropy over the cycle. In the adiabatic stages ($B \rightarrow C$ and $D \rightarrow A$), no heat flows, so $\Delta S_{BC} = \Delta S_{DA} = 0$. Over a full reversible cycle, the gas returns to its original state, so its entropy change is zero:

$$\Delta S = \frac{Q_H}{T_H} + 0 - \frac{Q_C}{T_C} + 0 = 0 \Rightarrow \frac{Q_C}{Q_H} = \frac{T_C}{T_H}$$

Step 4 — Substitute into the efficiency.

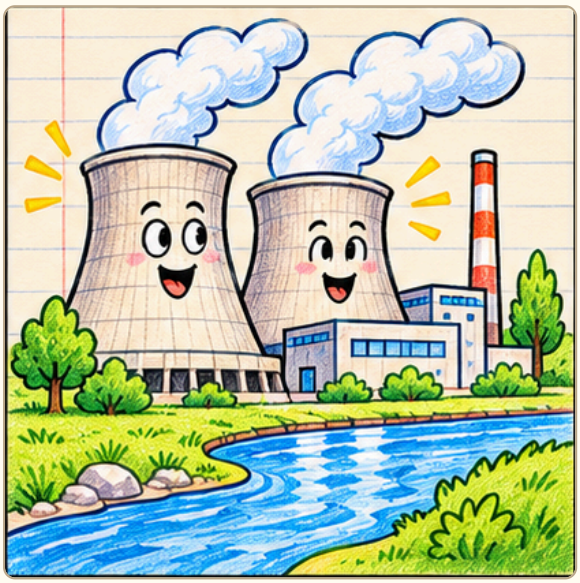
$$\eta = 1 - \frac{Q_C}{Q_H} \Rightarrow \eta_C = 1 - \frac{T_C}{T_H}$$

No engine working between these two temperatures can exceed this efficiency.

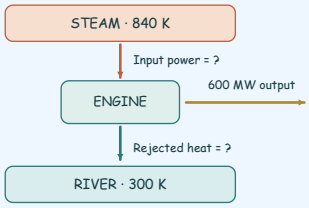
WORKED PROBLEM 13 / 13

Problem 13 — Power Station Efficiency and Entropy

Tsokos (CUP) B.4 · Carnot efficiency & second law · replaces SME: $\frac{500K}{300K}$ engineer's claim



QUESTION DATA & MODEL



Visual setup · Power Station Efficiency and Entropy

AI-generated notebook illustration · diagram geometry and question data define the physics.

A coal power station uses steam at 840 K and rejects heat to a river at 300 K. It delivers 600 MW of electrical power at an overall efficiency of 38%.

(a) Calculate the Carnot efficiency for these temperatures.

(b) Calculate the rate at which heat is rejected into the river.

(c) Calculate the rate at which the total entropy of the Universe increases because of this station.

(d) An investor is told about a new engine, working between the same two temperatures, with an efficiency of 70%. Using (a), advise the investor.

▼ Worked solution

(a) $\eta_C = 1 - 300/840 = 0.64$ (64%)

(b) Q_H per second $\frac{= 600}{0.38} = 1579 MW$ Q_C per second $= 1579 - 600 = 979 MW$ (more than the electrical output)

$$= \frac{-Q_H}{T_H} + \frac{Q_C}{T_C}$$

(c) ΔS per second $= \frac{-(1579 \times 10^6)}{840} + \frac{(979 \times 10^6)}{300}$ Positive, as the Second Law demands. A perfect

$$= -1.88 \times 10^6 + 3.26 \times 10^6$$

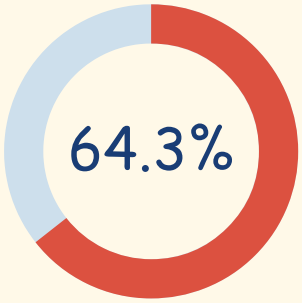
$$= +1.38 \times 10^6 \, JK^{-1} s^{-1}$$

Carnot engine would give exactly zero.

(d) 70% is above the 64% Carnot limit. It would mean $\frac{Q_C}{T_C} < \frac{Q_H}{T_H}$ — the Universe's entropy would decrease. That breaks the Second Law (the Kelvin and Clausius statements). Don't invest — the engine is impossible.

INTERACTIVE ARTIFACT 05 · CARNOT LIMIT

Explore the ceiling on efficiency.



Maximum reversible efficiency

Hot reservoir T_H / K

840 K

Cold reservoir T_C / K

300 K

η_C = 1 − 300 / 840 = 64.3%

Uses kelvin. This is the ideal upper limit, not the actual efficiency of a real engine.

10 10. EXTENSION: MAXWELL'S DEMON AND HEAT DEATH

FIGURE 29

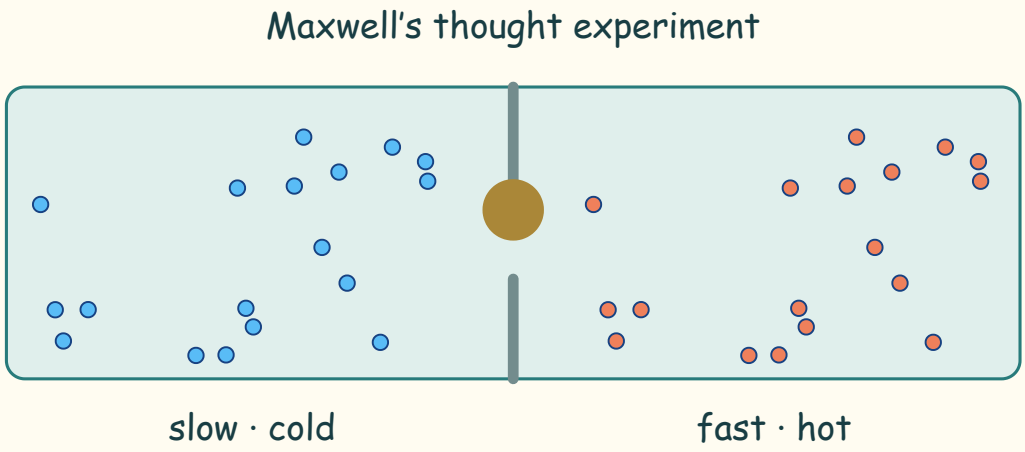


Figure 29. A tiny "demon" sitting at a trapdoor between two boxes of gas, letting fast molecules go right and slow ones go left.

In 1867, Maxwell proposed a thought experiment: a "demon" opens a trapdoor to let fast molecules pass one way and slow ones the other, making one side hot and one cold without work, apparently decreasing entropy. The resolution: the demon must measure and store information about each molecule, and erasing that information produces at least as much entropy. The Second Law holds.

FIGURE 30

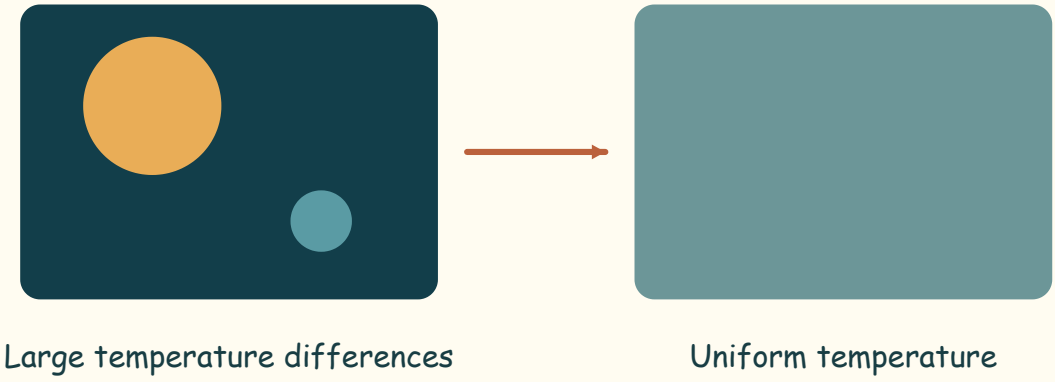


Figure 30. The universe drawn as a fading glow, every region at the same temperature — "heat death".

Kelvin and Clausius noted a consequence of the Second Law: if the entropy of the universe keeps increasing, it will eventually reach a uniform temperature. With no temperature difference, no heat engine could work. This is called the heat death of the universe. Summary: energy is conserved (First Law); total entropy increases (Second Law); and no engine can exceed the Carnot efficiency.

⭐ Bonus Trivia

Kelvin proposed his absolute temperature scale in 1848 — directly inspired by Carnot's work. That's why Carnot's formula needs kelvin, not Celsius. Rudolf Diesel disappeared from a ship crossing the English Channel in 1913. His body was found days later; his death is still a mystery. Your body, at rest, runs at roughly 100 W — about the same as an old filament bulb — and almost all of it ends up as heat, raising the entropy of the room. The Kelvin form of the Second Law means even a perfect power station must dump heat. That's why big power stations are built next to rivers or the sea, with giant cooling towers.

Question Sources Note

All 13 problems are newly written, at IB HL Paper 2 difficulty, in the style of Tsokos (Physics for the IB Diploma, Cambridge) and recent IB papers. The data used is standard: $R = 8.31\,\text{J mol}^{-1}\,\text{K}^{-1}$, $k_B = 1.38 \times 10^{-23}\,\text{J K}^{-1}$, L_f (ice) $= 3.34 \times 10^{5}\,\text{J kg}^{-1}$, L_v (water) $= 2.26 \times 10^{6}\,\text{J kg}^{-1}$. The isothermal work formula $W = nRT\ln(\frac{V_2}{V_1})$ is given in Problem 11, as it is not in the IB data booklet.